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**Development of a Digital Twin for Real-Time Monitoring
and Short-Term Forecasting of a Photovoltaic System**

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ABSTRACT

Solar photovoltaic (PV) systems have become one of the most popular modern renewable energy solutions. However, the performance of such PV systems depends on the changing environmental conditions, which include solar irradiance and temperature. Due to this, accurate power prediction is a concerning topic to deal with. Additionally, a lack of power output forecasting results in a significant loss due to the unexpected disruption of the power supply.

This project focuses on the development of a Digital Twin of a PV system installed at the Department of Electrical Engineering, Pulchowk Campus. The Digital Twin is designed in MATLAB/Simulink to represent the real system behavior under varying operating conditions. The parameters, such as irradiance and temperature, are the key inputs during the modeling and simulation process. For site-specific irradiance measurement, a low-cost Light-Dependent Resistor (LDR) based sensor was developed and calibrated to feed real-time data into the model.

A Seasonal Auto-Regressive Integrated Moving Average (SARIMA) model is applied to the historical irradiance and temperature data. The forecasted values are then propagated through the validated Simulink model to evaluate the system's predictive performance. With effective model training and validation, the digital twin can represent the physical system with reasonable accuracy.

The validation results indicate that the developed digital twin tracks measured power with $R^2 = 0.918$, while the SARIMA forecasting pipeline predicts daily energy yield within 0.5% of the reference value. The deviation between predicted and actual power can also be used as an early indicator for fault detection.

In general, the Digital Twin framework provides a practical approach to monitor, analyze, and predict the real-time operation of PV systems. This work reflects the potential of using simulation-based methods combined with data-driven forecasting tools to improve the reliability and efficiency of solar energy systems.

Keywords: Digital Twin, Photovoltaic System, SARIMA, Power Forecasting, Simulink, LDR Sensor

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Authors

TABLE OF CONTENTS

Copyright	i
Approval page	ii
Abstract	iii
Acknowledgements	iv
Table of contents	v
List of tables	viii
List of figures	x
List of acronyms and abbreviations	xi
 CHAPTER ONE: INTRODUCTION	 1
1.1. Background	1
1.2. Concept of Digital Twin	2
1.3. Machine Learning in Photovoltaic System	2
1.4. Problem Statement	3
1.5. Objectives	5
1.6. Scope of Project	5
1.7. Limitations of Project	6
1.8. Significance of Study	7
 CHAPTER TWO: LITERATURE REVIEW	 8
 CHAPTER THREE: RESEARCH METHODOLOGY	 12
3.1. System Architecture Overview	12
3.2. PV System Specifications	13
3.3. Simulink Model Development	14
3.3.1. PV Array Block Parameterisation	15
3.3.2. Boost Converter and MPPT Design	17
3.3.3. Saturation Tuning	18
3.3.4. Grid-tied Inverter and Control Loop	19
3.4. Data Acquisition Methods	19

3.4.1.	Real Power Measurement Setup	19
3.4.2.	LDR Irradiance Measurement	20
3.4.3.	Solcast Satellite-derived Irradiance Retrieval	20
3.4.4.	Temperature Data.....	21
3.5.	LDR Sensor Calibration.....	21
3.6.	Forecasting Methodology	22
3.6.1.	Choice of Forecasting Model	22
3.6.2.	SARIMA Model Configuration	23
3.6.3.	Training and Validation Strategy	23
3.7.	Validation Metrics.....	24
3.7.1.	Root Mean Square Error (RMSE)	24
3.7.2.	Mean Absolute Error (MAE)	24
3.7.3.	Mean Absolute Percentage Error (MAPE)	24
3.7.4.	Pearson Correlation Coefficient (r) and Coefficient of Determination (R-squared)	25
3.7.5.	Mean Bias Error (MBE)	25
3.7.6.	Energy Calculation	25
CHAPTER FOUR: RESULTS		27
4.1.	Overview	27
4.2.	Simulink Model Verification at Standard Test Conditions	27
4.3.	Field Observation Data	28
4.4.	Simulation of Power Output	29
4.5.	Validation Metrics.....	31
4.6.	Irradiance Band-wise Error Analysis	32
4.7.	Comparison of Irradiance Reference Sources	33
4.8.	Daily Energy Comparison	35
4.9.	Forecasting Performance	36
4.9.1.	Dataset and Forecast Configuration.....	36
4.9.2.	Forecast Accuracy of Irradiance and Temperature	37
4.9.3.	End-to-End Power Forecast.....	39

4.10.	Summary of Validation Results	40
4.11.	LDR Sensor Calibration Results	41
4.11.1.	Calibration Procedure	42
4.11.2.	Calibration Results	42
4.12.	Real-Time Operation Demonstration	43
4.12.1.	System Configuration	44
4.12.2.	Demonstration Results	44
CHAPTER FIVE: DISCUSSION		46
5.1.	Interpretation of Validation Results	46
5.2.	Comparison of Irradiance Reference Methods	47
5.3.	Structure of the Residual Errors	47
5.3.1.	Measurement-Side Error at Low Irradiance	48
5.3.2.	Model-Side Error at High Irradiance	49
5.3.3.	Asymmetric Error Pattern Explanation	49
5.4.	Forecasting Pipeline Performance and Limitations	50
5.5.	Practical Implications for PV Monitoring at Pulchowk	51
CHAPTER SIX: CONCLUSION AND FUTURE WORK		53
6.1.	Summary of Key Works	53
6.2.	Conclusions	54
6.3.	Recommendations for Future Work	55
REFERENCES		56
APPENDICES		59
APPENDIX A: Raw Observation Data		59
APPENDIX B: MATLAB Batch Simulation Script		62
APPENDIX C: SARIMA Forecasting Script		64
APPENDIX D: Arduino Code for LDR Sensor		72
APPENDIX E: PV Module Datasheet		76
APPENDIX F: MATLAB P&O MPPT CONTROLLER CODE		77

LIST OF TABLES

Table 3.1	Electrical and Physical Specifications of the PV Module.	14
Table 3.2	Internally Computed Single-Diode Model Parameters.	17
Table 3.3	P&O MPPT Algorithm Parameters.	18
Table 4.1	PV Module Parameters at STC (RenewSys DESERV 3M6H-335)	27
Table 4.2	Summary of Field Observations	29
Table 4.3	Validation Metrics for Simulated vs Measured Power	31
Table 4.4	Mean Simulation Error by Irradiance Band	32
Table 4.5	Energy Generated During Each Observation Window	35
Table 4.6	Forecast Accuracy Metrics for Day 8 (22 March 2020)	37
Table 4.7	End-to-End Forecast Pipeline Metrics for Day 8 Power Output	40
Table 7.1	Day 1 raw observations (2 April 2026, clear sunny).	59
Table 7.2	Day 2 raw observations (16 March 2026, partly cloudy).	60
Table 7.3	LDR calibration measurements used for irradiance estimation.	61
Table 7.4	RenewSys DESERV 3M6H-335 module specifications.	76

LIST OF FIGURES

Figure 3.1	Conceptual architecture of the developed PV Digital Twin system	12
Figure 3.2	Complete Simulink model block diagram	16
Figure 4.1	I–V and P–V characteristics of the PV module under STC	28
Figure 4.2	Time-series comparison of measured and simulated PV power output for (top) Day 1: 2 April 2026 (clear) and (bottom) Day 2: 16 March 2026 (partly cloudy)	30
Figure 4.3	Scatter of simulated versus measured power. (Left) LDR-based simulation, $R^2 = 0.918$. (Right) Solcast-based simulation, $R^2 = 0.703$. The dashed line represents the ideal 1:1 relationship.	32
Figure 4.4	Mean simulation error by irradiance band.	33
Figure 4.5	Simulation error percentage versus input irradiance for individual readings. The dashed lines indicate quadratic trend fits.	34
Figure 4.6	Comparison of LDR-based (point-source) and Solcast satellite-derived irradiance values for both observation days.	34
Figure 4.7	Energy generated during each observation window: measured versus simulated using LDR and Solcast irradiance.	35
Figure 4.8	SARIMA forecast validation. (Top) Hourly irradiance over Days 1–7 and forecast vs actual on Day 8 with 95 % confidence interval shaded. (Middle) Hourly temperature with the same structure. (Bottom) Detailed comparison of actual and forecasted irradiance and temperature on Day 8.	38
Figure 4.9	Day 8 PV power forecast. The reference power (blue) is the Simulink output when actual irradiance and temperature are supplied as inputs; the forecast power (orange dashed) is the Simulink output when SARIMA-forecasted inputs are supplied.	39
Figure 4.10	LDR voltage versus reference irradiance with fitted power-law calibration curve.	43

Figure 4.11 Real-time digital twin operation. (Top) Live irradiance measured by the calibrated LDR sensor. (Middle) Live ambient temperature from the temperature sensor. (Bottom) Simulated PV power output from the Simulink digital twin, computed in real-time from the live sensor inputs. 45

LIST OF ACRONYMS AND ABBREVIATIONS

ADC	:	Analog-to-Digital Converter
AM	:	Air Mass
ARIMA	:	Auto-Regressive Integrated Moving Average
CI	:	Confidence Interval
DC	:	Direct Current
DT	:	Digital Twin
GHI	:	Global Horizontal Irradiance
IEA	:	International Energy Agency
IGBT	:	Insulated-Gate Bipolar Transistor
IoE	:	Institute of Engineering
LDR	:	Light-Dependent Resistor
MAE	:	Mean Absolute Error
MAPE	:	Mean Absolute Percentage Error
MBE	:	Mean Bias Error
MPP	:	Maximum Power Point
MPPT	:	Maximum Power Point Tracking
NOCT	:	Nominal Operating Cell Temperature
P&O	:	Perturb and Observe
PI	:	Proportional-Integral
PV	:	Photovoltaic
PWM	:	Pulse Width Modulation

RMSE	:	Root Mean Square Error
SARIMA	:	Seasonal Auto-Regressive Integrated Moving Average
STC	:	Standard Test Conditions
V_{mp}	:	Voltage at Maximum Power Point
V_{oc}	:	Open-Circuit Voltage
I_{mp}	:	Current at Maximum Power Point
I_{sc}	:	Short-Circuit Current
ZCD	:	Zero-Crossing Detector

CHAPTER ONE: INTRODUCTION

1.1. Background

Fossil fuels constitute about 80% of global energy production and are limited in quantity. The rapid depletion of these energy sources and concerns over environmental pollution have shifted global attention toward increasing renewable energy generation alternatives. As per the International Energy Agency (IEA), renewable energy generation must account for over 90% of new power plants globally by 2030 in order to achieve net-zero emissions targets International Energy Agency, 2023. This has resulted in the rapid development of renewable sources for power generation globally.

Solar photovoltaic systems have emerged as one of the fastest-growing renewable energy technologies. Developments in the field of semiconductors have reduced the cost of electricity generation from solar by more than 90%, making it one of the most popular power generation systems International Renewable Energy Agency (IRENA), 2023. The total installed capacity of photovoltaic systems has already exceeded 1 terawatt globally and is projected to reach 5 terawatts by 2030, as renewable energy generation across Asia, Europe, North America, and Africa continues to grow Solar Power Europe, 2023.

The rapid growth of solar photovoltaic systems has introduced operational challenges. The power output from a solar PV system is primarily dependent on solar irradiance and temperature. Different studies show that output power can differ from the expected value due to factors such as cell temperature, shading patterns, soiling, and component aging. These issues can reduce energy production by 10% to 25% Dhimish et al., 2018.

These challenges in photovoltaic systems provide an opportunity to develop monitoring, modelling, and control frameworks capable of bridging the gap between the rated and actual performance of PV systems. The development of artificial intel-

ligence and sensor technologies helps address this gap and can result in optimum energy production.

1.2. Concept of Digital Twin

The concept of the Digital Twin was introduced by Dr. Michael Grieves at the University of Michigan in 2003. Since then, a number of scholars have contributed extensive literature across different databases. In recent years, the study of digital twins has witnessed a meteoric increase in popularity Yu et al., [2025](#).

A Digital Twin is a virtual replica of a physical system. In other words, a Digital Twin is defined as a dynamic, real-time digital representation of a physical system that is capable of representing its characteristics based primarily on physics-based equations, historical data, and sensor inputs. The digital twin has demonstrated significant potential across various industries, including both industrial and academic sectors, to increase productivity Yu et al., [2025](#). In the power and energy sector, Digital Twins are being deployed for wind turbine condition monitoring, transformer health assessment, and photovoltaic system management. The ability of a Digital Twin to provide continuous, impactful insight into system performance without requiring physical intervention makes it particularly well suited to distributed energy assets such as rooftop and utility-scale PV installations. Also, the use of a Digital Twin allows operators to monitor, improve efficiency, and enhance the productivity of PV systems in remote areas from urban centers.

1.3. Machine Learning in Photovoltaic System

Machine learning is a branch of artificial intelligence that helps in developing models and algorithms, allowing systems to learn patterns from input data and input-output relationships without being explicitly programmed for each task. The development and use of ML have achieved transformative results across different fields, ranging from academic research to medical diagnosis. In the energy sector, ML addresses problems such as load forecasting, fault diagnosis, energy management, and renewable energy optimization Gaviria et al., [2022](#).

Advanced control techniques are needed to ensure the reliable integration of PV systems into the grid, and this challenge can be addressed through the application of ML, particularly in solar PV systems. The relationship between environmental inputs (irradiance, temperature, shading) and electrical outputs (current, voltage, power) is highly nonlinear and difficult to capture accurately using analytical models alone Yu et al., 2025 Gaviria et al., 2022. ML models such as the Autoregressive Integrated Moving Average (ARIMA) model are capable of learning these complex relationships directly from data. ML also enables the forecasting of future system behavior based on historical patterns, which is important for proactive maintenance and energy planning.

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model employed in this project is an application of machine learning used for energy forecasting, which is challenging in solar PV systems. The training process using real-time and historical data minimizes prediction error, helping ML approximate the input-output relationships present in the training dataset and simulate power output by predicting irradiance values.

1.4. Problem Statement

The increased adoption of solar energy has created an important opportunity for clean and economical power generation. However, this growth has also introduced critical operational challenges, as solar power is intermittent and varies with changing environmental conditions. Unlike conventional systems, in which energy can be generated on demand, PV systems produce electricity as a direct function of solar irradiance, ambient temperature, and atmospheric conditions. These parameters fluctuate continuously and are subject to rapid, unpredictable changes driven by multiple environmental factors Antonanzas et al., 2016. As a result, PV systems become less predictable sources of electrical generation, and accurate, reliable power forecasting becomes essential for their effective integration into modern electrical grids Quan et al., 2014.

A number of forecasting techniques had been proposed by different researchers to

predict the output of solar PV systems. The persistence model assumes that the next irradiance value is equal to the most recently observed value whereas probabilistic methods for irradiance prediction may still produce deviations beyond the prescribed limit. Likewise, moving-average-based methods are simple to implement but often fail to capture varying temporal patterns and seasonal variations. These approaches are therefore insufficient for short-term forecasting and may lead to larger errors. The Autoregressive Integrated Moving Average (ARIMA) model addresses some of these limitations by capturing autoregressive relationships, applying integration to remove non-stationary components, and incorporating moving average terms to smooth the differenced series. As a result, ARIMA provides a statistically reliable and widely adopted approach for energy-demand estimation and short-term solar irradiance forecasting Sharma and Mishra, 2023.

MATLAB/Simulink and PVSyst are widely employed for the simulation of solar PV systems and for modelling their electrical characteristics. These models enable detailed representations of PV modules through equivalent circuit models and corresponding mathematical equations. The problem in conventional practice is that these models operate on manually entered assumptions rather than dynamically predicted real-world values. This static input approach creates a considerable gap between the simulated and actual electrical performance of the system, along with delays caused by manually entering predicted values.

The noticeable advancement in both time-series forecasting and PV system simulation is beneficial for power prediction. However, there remains a gap in developing an integrated system that connects power forecasting driven by irradiance prediction directly with physics-based PV simulation in Simulink. This project addresses this gap by developing an SARIMA-based short-term forecasting model driven by historical data and coupling its predicted irradiance values as dynamic input to a Simulink model. This integration forms the basis of the Digital Twin concept proposed in this project and supports forecasting-driven operational planning under varying environmental conditions.

1.5. Objectives

- i. To develop a Digital Twin of the existing solar PV system at the Renewable Laboratory, Pulchowk Campus, incorporating SARIMA models driven by historical meteorological data for short-term irradiance and temperature forecasting and corresponding power output prediction.
- ii. To create a simple pyranometer using LDR, arduino, resistor capable of providing input of the irradiance value to simulink and also calibrating the developed pyranometer.
- iii. To compare the output power of physical solar PV system, simulated power output and predicted power output and help in optimizing electrical parameters.

1.6. Scope of Project

This project is focused on Development of Digital Twin of a Solar PV system in Renewable Lab under Department of Electrical Engineering, Pulchowk Campus, Lalitpur. The scope of the work is to study the electrical performance of the PV system and prediction of intermittent solar power. This allows reliable integration of Solar energy in the grid enabling load sharing and scheduled other sources operations.

The SARIMA model driven by historical data in this project is limited to short term irradiance forecasting which is used as an input to simulation model. The meteorological dataset used for model development and validation will be obtained from the meteorological station in the rooftop of Department of Electronics and Computer Engineering, Pulchowk Campus.

The simulink PV system developed is the digital copy of existing physical system in the renewable lab. The parameter of each component of the simulink model is standard manufacturer dataset of each equipment of the existing solar system representing the virtual replica of PV system in lab.

The entire work is implemented in Simulink, validated from the real PV system

and the deviation is observed and analysed. This study is limited to implementation of hardware for irradiance measurement only with all other works carried in MATLAB-Simulink. It does not extend its study for the disturbances and electrical parameters variations due to harmonics or penetration of PV system in grid.

1.7. Limitations of Project

The project presents a meaningful contribution towards short term forecasting and power system planning, however, several limitations are acknowledged which may deviate the result and leads to inaccuracy. First, the implementation of pyranometer for solar irradiance measurement using LDR, Arduino and resistance is calibrated using mobile irradiance measurement which could carry error. These errors can potentially deviate the simulated power from the real power output. Secondly, the ARIMA model is a linear time series forecasting method with its ability unsuitable to capture nonlinear and complex relationships between solar irradiance and influencing environmental variables such as cloud cover, pollution and rapid variables like humidity, wind speed, ambient temperature variations. Also, the accuracy of the ARIMA model is dependent on the quality, completeness and length of historical data.

The Simulink model developed in this study is based on mathematical equations however, in practice, PV module parameters may vary with ageing and temperature variations, which means the model may not fully represent the behaviour of a real system to some extent. The proposed Digital Twin is implemented entirely within a software simulation and does not incorporate real-time sensor data from a physical PV installation due to absence of sensors. Finally, the scope of this study is limited to short term irradiance forecasting and does not address medium-term or long-term forecasting, which would require different modelling approaches and are relevant for broader energy planning and grid management applications.

1.8. Significance of Study

The development of Digital Twin of a real system and incorporating the forecasting model possesses significant value across academic, technical, operational and grid planning system.

From an academic perspective, this study contributes to the growing research on energy forecasting of renewable energy generation driven by environmental constraints. Although ARIMA models are used by researchers for load forecasting, their application in energy forecasting remains relatively unexplored. This study provides a basic method of utilizing this model for energy forecasting that can be further extended by researchers for a lower-error forecasting system. Also, this study can be used as a base for implementing the forecasting system in other renewable generation sources such as wind.

From the technical perspective, the proposed work shows how a forecast-driven Digital Twin can bridge the gap between static PV simulation and real-world dynamic operating conditions. By replacing fixed irradiance based on assumptions in the Simulink model with ARIMA-predicted values, the simulation becomes a more realistic and practically useful tool for estimating PV electrical output under varying environmental conditions. Also, the implementation of a low-cost pyranometer as a sensor serves as a foundation for incorporating real-time sensors in the PV station, ensuring a closed-loop feedback mechanism.

From an operational perspective, short term energy forecasting provides a foundation for energy management and system planning. Accurate forecasting obtained with larger datasets can support load scheduling and reliable integration of the solar power system. Overall, this study supports the concept of advancing smart, data-driven, and sustainable energy systems.

CHAPTER TWO: LITERATURE REVIEW

The global push toward renewable energy has pushed up the use of photovoltaic (PV) systems in recent years. Their output, though, is not constant. It changes with the weather, with shading, with how old the components are, and with the limits of the converter used. All these things make it harder to predict and control how a PV system performs. This is where Digital Twin (DT) technology comes in. A DT is a virtual copy of the physical PV system that keeps updating itself using sensor data. Work done by Fong Mollineda and García (2024) and by Okeke et al. (2025) points out that DTs are useful for live monitoring, predictive maintenance, and improving day-to-day operation.

The digital twin idea actually started during NASA’s Apollo missions. Engineers built virtual models of the spacecraft and used them to simulate how the real one would behave. The same idea later moved into many other industries, including power systems. In PV applications today, a DT usually pulls together IoT devices, cloud platforms, and data analytics so that the real and virtual systems stay in sync. Alao et al. (2024) explains that DTs improve the efficiency and reliability of PV plants, but the work also flags some real challenges: cybersecurity risks, costs, and the lack of common communication standards. Massel et al. (2021) make a slightly different point – they show that engineers can use DTs to try out configuration changes in a safe virtual environment first, before changing anything in the actual hardware.

Before going deeper into how DTs are used, it helps to look at the PV module itself. A module is just many solar cells wired together, and they convert sunlight into electricity. The performance, though, depends on a lot of things at once: irradiance, temperature, shading, soiling, and so on Fong Mollineda and García, 2024 Okeke et al., 2025. To get the most out of the panel, Maximum Power Point Tracking (MPPT) methods are used. The Perturb and Observe (P&O) method

is the one most people start with because it is simple. Atallah et al. (2014) and Stein (2012) show how P&O works in practice – it nudges the operating point a little, watches how power changes, and adjusts to stay near the maximum.

How accurate a PV simulation or digital twin is depends a lot on the irradiance data we feed in. The literature shows several ways to measure irradiance, and each has its own trade-offs in cost, accuracy, and area covered. Reference-grade pyranometers are the most accurate. They are also expensive, which makes them hard to use in smaller projects. Satellite-based services like Solcast give wide coverage and are now used a lot for solar resource estimation, but their values are averaged over a large area. So they often miss quick local cloud changes Fong Mollineda and García, 2024. As a low-cost option, some studies have used Light-Dependent Resistors (LDRs) calibrated against reference instruments. The voltage from an LDR follows a power-law relation with irradiance. The sensor is not perfect – it loses sensitivity at very low light and also drifts a bit with temperature – but if the calibration is done carefully, it works fine for site-specific measurement Stein, 2012.

Forecasting PV output is another area that has been studied widely, mainly because of how important it is for grid operation and planning Antonanzas et al., 2016. The methods used can be split roughly into three groups: physical, statistical, and machine-learning. The simplest ones, like persistence and moving averages, are easy to set up but they cannot really capture daily patterns or sudden changes. Statistical models such as ARIMA and its seasonal version SARIMA do better because they handle autoregressive behavior, trend, and periodic patterns in the data Sharma and Mishra, 2023. SARIMA in particular suits solar irradiance well, since the data has a clear daily cycle. Newer studies have moved toward machine-learning techniques, like Artificial Neural Networks, Long Short-Term Memory networks, and gradient-boosted trees. These can model nonlinear behavior, but they need more training data and are harder to interpret Gaviria et al., 2022 Quan et al., 2014. For short-term forecasting at a single site, especially when historical data is limited, SARIMA is still a strong starting point. That is

why we have used it in this work.

Recent literature shows DTs being used to actually improve PV performance, not just monitor it. Karunanidhi et al. (2024) built a DT for a 3×3 PV array to deal with partial shading. They used schemes like Equal Irradiance Distribution and the Skyscraper Puzzle pattern, and reported about 40% more power output and a 20% better fill factor. Kolahi et al. (2024) took a different angle for large solar plants. Their Digital-PV platform brings together UAV simulations, Unreal Engine, AirSim, and deep-learning models to keep track of how the plant is operating. It also produces synthetic data that helps train AI-based monitoring systems.

When historical data is hard to get, physics-based DTs become especially useful. Zhang et al. (2025) built one that includes thermal modeling, wind-flow simulation, and the electrical side of the panel. Their results were strong: less than 1% error at module level and around 96% accuracy in real field tests. The advantage of this kind of approach is that it does not depend on long sensor histories. The model uses physical equations directly, so it can work even when there is not much past data to learn from.

Most PV digital twins in the literature fall into one of three groups: data-driven, physics-based, or hybrid Fong Mollineda and García, 2024 Zhang et al., 2025. Data-driven ones use machine-learning on past sensor data – they do well in steady conditions but struggle when the weather changes suddenly. Physics-based models stick to physical laws, so they hold up better in extreme or rare situations. Hybrid models try to take the good parts of both, and they usually end up with better accuracy and flexibility.

Looking across the literature, two things stand out. First, most reported PV digital twins are either fully physics-based or fully data-driven – not many combine both with a forecasting layer. Second, when forecasting models are tested, they are usually validated on their own, not as part of a digital twin pipeline. So there is a clear gap in work that takes a forecasted irradiance and temperature signal, runs it through a validated physics-based digital twin, and produces a short-term

power forecast for a specific site. On top of that, very few academic studies pair such a pipeline with a low-cost irradiance sensor placed right at the panel. Our work tries to fill these gaps. We use a calibrated LDR sensor for site-specific measurement, a SARIMA model to forecast irradiance and temperature, and a validated Simulink digital twin that turns those forecasts into day-ahead power predictions.

Overall, what the literature shows is that DT technology really does help in predicting PV performance, optimizing MPPT, and supporting operational decisions. There are still problems to solve – security concerns, high implementation cost, and the lack of common standards. But the research as a whole keeps pointing the same way: DT frameworks are being adopted more and more, both at the small scale and at the utility scale.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1. System Architecture Overview

The developed Digital Twin framework is classified into three integrated layers: Physical, Virtual, and Data & Forecasting, enabling continuous interaction between the real PV system and its digital replica.

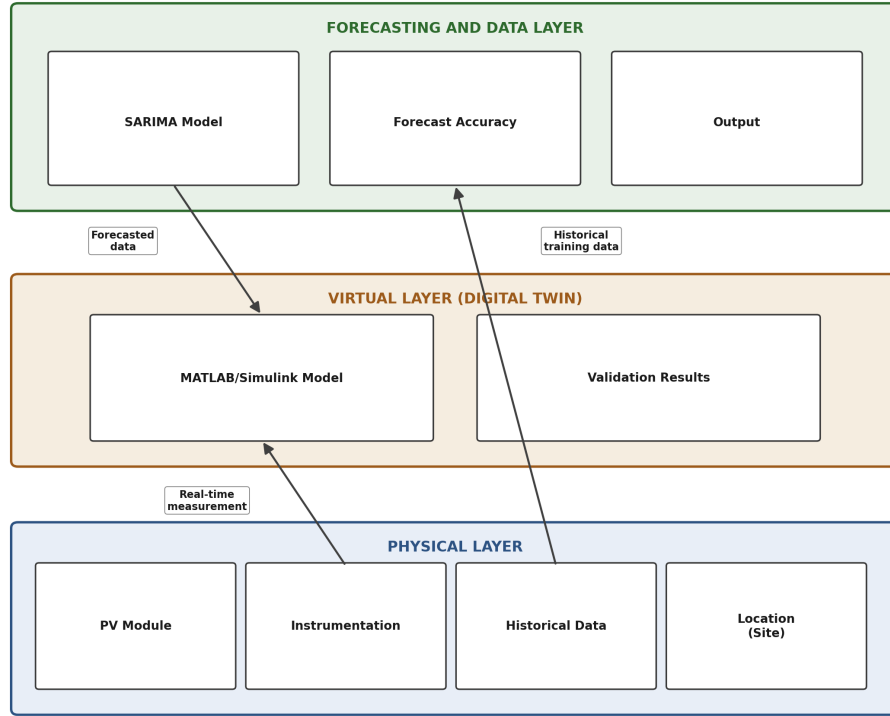


Figure 3.1 Conceptual architecture of the developed PV Digital Twin system

The **Physical layer** consists of a solar PV system installed at the Department of Electrical Engineering, Pulchowk Campus. Solar irradiance incident on the RenewSys 335 W module drives a DC–DC boost converter with a Perturb and Observe (P&O) MPPT algorithm and a single-phase grid-tied inverter, supplying AC power to the grid. Real-time data of irradiance, temperature, and power output are recorded between 11:00 and 15:00 as ground truth measurements.

The **Virtual Layer** is developed in MATLAB/Simulink using Simscape Electrical. It models the PV system based on datasheet parameters and uses irradiance and temperature as inputs to generate simulated power output. The model includes a single-diode PV model, boost converter, MPPT controller, and PWM inverter.

The **Data and Forecasting Layer** uses Seasonal AutoRegressive Integrated Moving Average (SARIMA) models to learn from past data and predict short-term values of solar irradiance and temperature. These forecasted environmental inputs are then supplied to the Virtual Layer to generate the corresponding power forecast. The system works in a connected way: real measurements from the Physical Layer are used to validate the Virtual Layer, while the forecast results are also fed into the Virtual Layer to simulate and test future system behavior.

3.2. PV System Specifications

The PV module used in this work is a RenewSys DESERV 3M6H-335 multicrystalline silicon module manufactured by RenewSys India Pvt. Ltd. It has a rated capacity of 335 W under Standard Test Conditions (STC: irradiance of 1000 W/m^2 , cell temperature of 25°C , and air mass AM 1.5). The module is certified under IEC 61215, IEC 61730, IEC 61701, and IEC 62716 standards, and comes with a manufacturer's power tolerance of 0 to $+4.99 W_p$. The main electrical and physical parameters, taken from the datasheet and nameplate, are presented in Table 3.1.

Table 3.1 Electrical and Physical Specifications of the PV Module.

Parameter	Symbol	Value
Rated maximum power	$P_{\max}$	335 W
Open-circuit voltage	V_{oc}	46.27 V
Short-circuit current	I_{sc}	9.41 A
Voltage at maximum power point	V_{mp}	37.9 V
Current at maximum power point	I_{mp}	8.85 A
Number of cells per module	N_{cell}	72
Cell type	—	Multi-crystalline Si
Maximum system voltage	—	1500 V
Temperature coefficient of V_{oc}	β	$-0.36099\%/^{\circ}\text{C}$
Temperature coefficient of I_{sc}	α	$+0.102\%/^{\circ}\text{C}$
Module weight	—	21.2 kg
Output tolerance	—	0 to +3 %

The module is mounted at a fixed tilt at the Department of Electrical Engineering, Pulchowk Campus (latitude 27.6588°N, longitude 85.3247°E, altitude 1350 m above sea level). The system is integrated with a grid-tied power conditioning unit, which delivers the converted AC power to the local distribution network.

3.3. Simulink Model Development

The PV system was designed and simulated in MATLAB/Simulink using the Simscape Electrical (SimPowerSystems) library. The model includes all the main components needed for grid-connected operation, such as the PV array, DC–DC boost converter, Perturb and Observe (P&O) MPPT controller, single-phase voltage source inverter, output filter, and a grid voltage source representing the local utility supply.

All these components are connected in a way that the model can represent how solar energy is converted into AC power and delivered to the grid. This setup

makes it easier to study how the system behaves under different conditions without building the physical system, which helps in better understanding and improving the overall performance.

The block diagram of the developed Simulink model is shown in Figure 3.2.

3.3.1. PV Array Block Parameterisation

The PV array was modelled using the standard PV Array block from the Simscape Electrical library. This block is based on the single-diode equivalent model, which is widely used because it represents the actual behaviour of a solar cell quite well without making the model too complicated.

The output current of the PV cell is given by:

$$I = I_{ph} - I_0 \left(\exp\left(\frac{V + IR_s}{nV_t}\right) - 1 \right) - \frac{V + IR_s}{R_{sh}} \quad \text{Equation 3.1}$$

In this equation, I_{ph} is the current generated due to sunlight, and I_0 is the diode saturation current. The voltage across the cell is represented by V . Here, R_s is the series resistance, which accounts for internal losses, while R_{sh} is the shunt resistance, representing leakage paths inside the cell. The parameter n is the diode ideality factor, and V_t is the thermal voltage.

For modelling, the PV block was set to user-defined mode, where key datasheet values like V_{oc} , I_{sc} , V_{mp} , and I_{mp} were entered. Based on these values, the block internally calculates the remaining parameters so that the simulated characteristics match the real PV module as closely as possible.

The calculated parameters are summarized in Table 3.2.

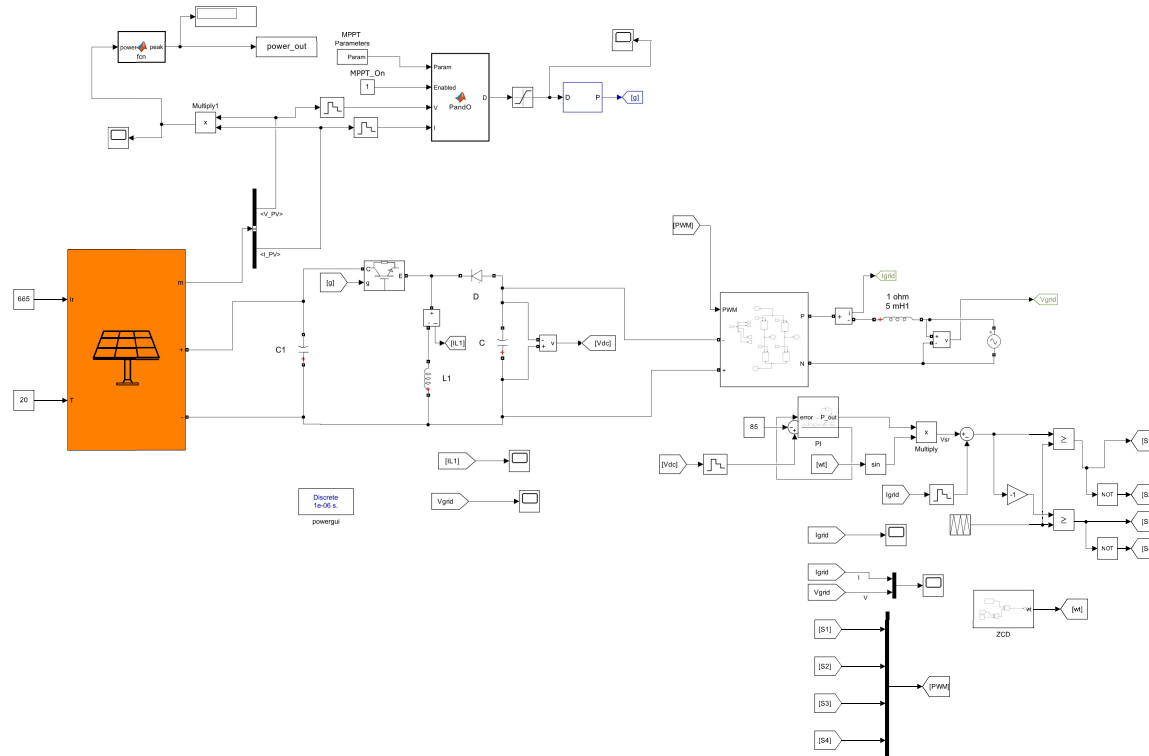


Figure 3.2 Complete Simulink model block diagram

Table 3.2 Internally Computed Single-Diode Model Parameters.

Parameter	Symbol	Value
Light-generated current	I_{ph}	9.416 A
Diode saturation current	I_0	3.509×10^{-10} A
Diode ideality factor	n	1.0421
Series resistance	R_s	0.29969Ω
Shunt resistance	R_{sh}	471.1865Ω

The PV array block takes solar irradiance and temperature as inputs, denoted by G_{in} (W/m²) and T_{in} (°C). Based on these, it produces the output voltage V_{PV} and current I_{PV} , and their product gives the power output of the PV array. In simple terms, this setup lets the model behave very close to a real solar panel, making it easier to study how changes in sunlight and temperature affect the overall performance.

3.3.2. Boost Converter and MPPT Design

A DC–DC boost converter was used in the system to increase the PV panel voltage to the required DC link level for the inverter. The converter is made up of an input capacitor C_1 , an inductor L_1 , an IGBT switch, a diode, and an output capacitor C . Together, these components help in stepping up the voltage and keeping the output stable.

The output voltage of the boost converter is given by:

$$V_{out} = \frac{V_{in}}{1 - D} \quad \text{Equation 3.2}$$

Here, D is the duty cycle of the IGBT switch ($0 \leq D < 1$). By changing this duty cycle, the converter controls how much the input voltage is increased at the output.

For extracting maximum power from the PV array, the Perturb and Observe

(P&O) MPPT algorithm was used, implemented through a function block. The idea is simple: the reference voltage $V_{dc,ref}$ is slightly varied, and the change in output power is observed. If the power increases, the change continues in the same direction; if it decreases, the direction is reversed. This way, the system keeps operating close to the maximum power point.

The parameters used for the P&O MPPT algorithm are given in Table 3.3.

Table 3.3 P&O MPPT Algorithm Parameters.

Parameter	Value
Initial reference voltage ($V_{dc,ref}$)	30 V
Voltage perturbation step (ΔV)	0.1 V
Upper limit (V_{max})	100 V
Lower limit (V_{min})	0 V
Sample time	2 μs

3.3.3. Saturation Tuning

To ensure safe and stable operation of the boost converter, a saturation block was placed between the MPPT controller output and the duty cycle input. The main idea behind this is to keep the duty cycle within a safe range so that the converter operates smoothly and does not go into unstable conditions.

When the duty cycle gets very close to 1, the boost converter gain becomes highly nonlinear, and the output voltage increases very rapidly. In such a case, the MPPT can become unstable and sometimes even show numerical issues during simulation. To avoid this, the upper limit of the saturation block was tuned manually by testing different values and checking the steady-state power output of the PV panel under standard test conditions (STC).

Finally, a value of 0.87 was chosen as the upper limit because it gave a stable output power of around 335 W at the PV terminals, which matches the rated power of the panel. The lower limit was kept at 0 so that the duty cycle never becomes negative, which is not physically possible.

3.3.4. Grid-tied Inverter and Control Loop

A single-phase voltage source inverter was used to convert the DC output from the boost converter into AC and feed it into the grid while keeping it synchronized with the local supply. The switching of the inverter is handled using Pulse Width Modulation (PWM), and the control signals are generated through a structure made of an inner current control loop and an outer voltage control loop.

On the grid side, a filter is used to improve power quality. It consists of a 5 mH inductor and a 1 Ω damping resistor, which together help reduce harmonic content in the injected current. For proper control and stability, both the grid voltage and grid current at the point of common coupling are continuously measured and fed back into the controller. In the simulation, the grid is represented using an ideal single-phase AC voltage source with rated parameters to reflect actual grid conditions as closely as possible.

3.4. Data Acquisition Methods

To check whether the Simulink model is actually behaving like the real system, and also to prepare data for the forecasting part of the study, measurements were taken from the installed PV system. The data was collected on two separate days, chosen on purpose so that different weather conditions could be captured and compared.

During these observation periods, readings were taken directly from the system using the available sensors and measuring instruments. This helped in recording the actual operating behaviour of the PV system in real conditions, rather than relying only on simulation results.

3.4.1. Real Power Measurement Setup

The instantaneous DC power from the PV module was obtained from the existing power conditioning unit monitoring interface, which displays voltage, current, and power in real-time. The readings were noted manually at 10-minute intervals from

11:00 to 15:00 on each observation day. This period was selected to capture peak solar generation and keep the dataset practical for analysis.

3.4.2. LDR Irradiance Measurement

To estimate irradiance at the panel location, a mobile application was used which relies on the phone’s ambient light sensor. It provides irradiance values in W/m^2 , giving a quick and practical estimate of the sunlight available at the site.

For each reading, the phone was kept flat on the PV panel so that it faces the same light conditions as the module itself. The irradiance readings were taken together with power and temperature measurements to keep everything properly matched.

This method is quite easy to use in the field, but it is not highly precise since smartphone light sensors are mainly designed for adjusting screen brightness rather than scientific measurements. Because of this, slight variations in readings are expected.

Still, it is useful for observing real, local changes in sunlight, especially sudden fluctuations caused by passing clouds. These short-term variations are often not captured well in satellite data, so this method helps in understanding actual site conditions better. The limitations are further discussed in the Sources of Error section in Chapter 5.

3.4.3. Solcast Satellite-derived Irradiance Retrieval

For comparison with the on-site measurements, irradiance data was also taken from Solcast Solcast, [2020](#), a commercial solar data provider that estimates solar radiation using geostationary satellite imagery combined with atmospheric models to calculate global horizontal irradiance. For the location of Pulchowk Campus, irradiance data for the selected observation days was collected and then interpolated so that it aligns with the 10-minute time steps used in the study.

This dataset helps in observing the overall solar pattern during the day and pro-

vides a reference for how irradiance varies over time under regional weather conditions.

Unlike the mobile measurements, these values are not point-specific and instead represent average irradiance over a wider geographical area around the site. This makes them useful for comparison, but they may slightly differ from actual conditions at the PV panel location.

3.4.4. Temperature Data

Ambient temperature values for each time stamp were collected from local meteorological records and cross-checked with the readings shown on the on-site weather indicator to make sure the data stayed consistent during the measurement period.

In this work, the cell temperature was not measured directly because of practical constraints. Instead, it is handled within the Simulink PV array block, where ambient temperature along with solar irradiance is used to internally estimate the thermal condition of the PV module. This gives a reasonably realistic representation of how the panel behaves in actual operating conditions without needing separate physical sensors for cell temperature.

3.5. LDR Sensor Calibration

The sensor is placed right next to the solar panel so that it can measure the sunlight hitting the panel directly. Satellite data can be less accurate for location conditions and the phone sensors provide inaccurate measurements. The best cost-effective alternative is an LDR. An LDR is a passive component whose resistance decreases nonlinearly with increasing incident light intensity. The LDR is connected in a resistive voltage-divider configuration with a fixed resistor and powered by an Arduino Uno 5 V supply. The voltage at the divider midpoint was sampled by the Arduino's analog input and recorded synchronously with simultaneous mobile app irradiance readings.

The power law for the voltage-irradiance relationship of an LDR can be found as

$$G = a \cdot V^b.$$

where G is irradiance in W/m^2 , V is the LDR voltage output in volts, and a and b are calibration coefficients to be determined experimentally.

While calibrating the LDR against the mobile application, the limitation of the mobile application sensor transfers its inaccuracy to the LDR producing a non-accurate irradiance reference. A more careful calibration would require a certified pyranometer or a self-reference method based on the panel's short-circuit current. These possible calibration approaches are discussed in the Recommendations for Future Work in Chapter 6.

3.6. Forecasting Methodology

3.6.1. Choice of Forecasting Model

For short-term forecasting of solar irradiance and temperature, the Seasonal Autoregressive Integrated Moving Average (SARIMA) model was selected. SARIMA was chosen because both irradiance and temperature exhibit strong daily seasonal patterns driven by the diurnal solar cycle, which the seasonal differencing terms in SARIMA can capture directly. A pure ARIMA model would not adequately represent the daily periodicity present in the data.

A two-stage forecasting approach is adopted in this work. In the first stage, separate SARIMA models are trained for irradiance and temperature using historical data, and forecasts are produced for the next day. In the second stage, the forecasted irradiance and temperature values are supplied as inputs to the validated Simulink digital twin to produce the corresponding hourly power forecast. This separates the statistical forecasting of environmental drivers from the physics-based modelling of the panel response, allowing each component to be validated independently.

3.6.2. SARIMA Model Configuration

The SARIMA model is a generalisation of the ARIMA model that incorporates seasonal differencing and seasonal moving-average terms. It is described by the notation $\text{SARIMA}(p, d, q) \times (P, D, Q)_S$, where:

- p, d, q are the non-seasonal autoregressive, differencing, and moving average orders;
- P, D, Q are the corresponding seasonal autoregressive, differencing, and moving average orders;
- S is the seasonal period (number of samples per seasonal cycle).

The general form of the SARIMA model is given by Equation 3.3:

$$\Phi_P(B^S)\phi_p(B)(1-B)^d(1-B^S)^D y_t = \Theta_Q(B^S)\theta_q(B)\varepsilon_t \quad \text{Equation 3.3}$$

where B is the backshift operator, $\phi_p(B)$ and $\theta_q(B)$ are the non-seasonal AR and MA polynomials, $\Phi_P(B^S)$ and $\Theta_Q(B^S)$ are the seasonal AR and MA polynomials, and ε_t is the white noise residual.

For the forecasting task in this project, separate SARIMA models were configured as $\text{SARIMA}(1, 1, 1) \times (1, 1, 1)_{24}$ for irradiance and temperature. The seasonal period $S = 24$ corresponds to one full diurnal cycle for hourly data (24 samples per day). The dependent variable y_t represents the hourly irradiance in W/m^2 for the irradiance model and ambient temperature in $^\circ\text{C}$ for the temperature model.

3.6.3. Training and Validation Strategy

The forecasting dataset contains eight days of historical hourly data, which gives a total of 192 observations. From this, the first seven days (168 observations) were used for training the SARIMA model. The last day (24 observations) was kept for validation to test the model performance on unseen data. The SARIMA

models for irradiance and temperature were trained independently using the same training set.

3.7. Validation Metrics

To evaluate how close the simulated power values are to the measured power values, and also to check how accurate the forecasted values are compared to the actual data, five different error metrics were used in this study. In the equations below, y_i represents the actual measured value, $\hat{y}_i$ represents the predicted or simulated value, and n represents the total number of samples used in the calculation.

3.7.1. Root Mean Square Error (RMSE)

RMSE is used to find the average size of the error between measured and predicted values. It gives more weight to larger errors, so bigger mistakes affect the result more. It is expressed in the same unit as the original data, so it is easy to interpret.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad \text{Equation 3.4}$$

3.7.2. Mean Absolute Error (MAE)

MAE is calculated by taking the average of the absolute difference between actual and predicted values. It gives a simple idea of how far the predictions are from the real values on average. Like RMSE, it is also in the same unit as the data.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad \text{Equation 3.5}$$

3.7.3. Mean Absolute Percentage Error (MAPE)

MAPE expresses the error in percentage form. This makes it easier to understand how large the error is compared to the actual value, and it is useful for comparing

different datasets.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad \text{Equation 3.6}$$

3.7.4. Pearson Correlation Coefficient (r) and Coefficient of Determination (R-squared)

The correlation coefficient r shows how strongly the predicted and actual values are related. If it is close to 1, it means both follow a similar pattern. The value R^2 is the square of r , and it shows how much of the variation in the data is explained by the model.

$$r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}}, \quad R^2 = r^2 \quad \text{Equation 3.7}$$

3.7.5. Mean Bias Error (MBE)

MBE is used to check whether the model is consistently overpredicting or underpredicting the actual values. If the value is positive, it means the model is generally giving higher values than the actual data.

$$MBE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad \text{Equation 3.8}$$

3.7.6. Energy Calculation

To calculate total energy over a time period, the trapezoidal rule is used on discrete power values. This gives a good approximation of the total energy generated during the sampling interval.

$$E = \sum_{i=1}^{n-1} \frac{P_i + P_{i+1}}{2} \Delta t \quad \text{Equation 3.9}$$

These metrics together give a complete understanding of how the model is performing. Some metrics focus on average error, some focus on relative error, and others help in understanding trend matching and whether the model is biased in one direction.

CHAPTER FOUR:RESULTS

4.1. Overview

This chapter presents the experimental and simulation outcomes of the developed PV digital twin system. The results are divided into various major sections as presented below.

The observations were conducted using a single RenewSys DESERV 3M6H-335 multi-crystalline silicon module (335 Wp rated), installed at the Institute of Engineering, Pulchowk Engineering College, Lalitpur, Nepal. Measurements were recorded between 11:00 AM and 3:00 PM to capture the peak solar generation.

4.2. Simulink Model Verification at Standard Test Conditions

Prior to analysing the real-world data, the Simulink PV array model was verified against the manufacturer's nameplate rating under Standard Test Conditions (STC: irradiance $G = 1000 \text{ W/m}^2$, cell temperature $T = 25^\circ\text{C}$, air mass AM 1.5). The model parameters were directly extracted from the rating plate, as summarised in Table 4.1.

Table 4.1 PV Module Parameters at STC (RenewSys DESERV 3M6H-335)

Parameter	Symbol	Value	Unit
Maximum power	$P_{\max}$	335.415	W
Open-circuit voltage	V_{oc}	46.27	V
Short-circuit current	I_{sc}	9.41	A
Voltage at MPP	V_{mp}	37.9	V
Current at MPP	I_{mp}	8.85	A
Cells per module	N_{cell}	72	—
Temperature coefficient of V_{oc}	β	-0.36099	%/ $^\circ\text{C}$
Temperature coefficient of I_{sc}	α	+0.102	%/ $^\circ\text{C}$
Series resistance	R_s	0.29969	Ω
Shunt resistance	R_{sh}	471.1865	Ω

To ensure stable Maximum Power Point Tracking (Section 3.3.2.), the boost con-

verter duty cycle saturation limit was set at 0.87. The model was simulated with $G = 1000 \text{ W/m}^2$ and $T = 25^\circ\text{C}$. The output power obtained was 335 W at steady state, which was close to the rated 335 Wp value.

The corresponding I–V and P–V characteristic curves generated by the PV array blocks are shown in Figure 4.1. It can be confirmed that the maximum power point occurs at $V_{\text{mp}} = 37.9 \text{ V}$ and $P_{\text{max}} = 335.415 \text{ W}$. Thus, these parameters were confirmed by comparing the manufacturer’s specifications under the specified conditions with the observed model’s outcome.

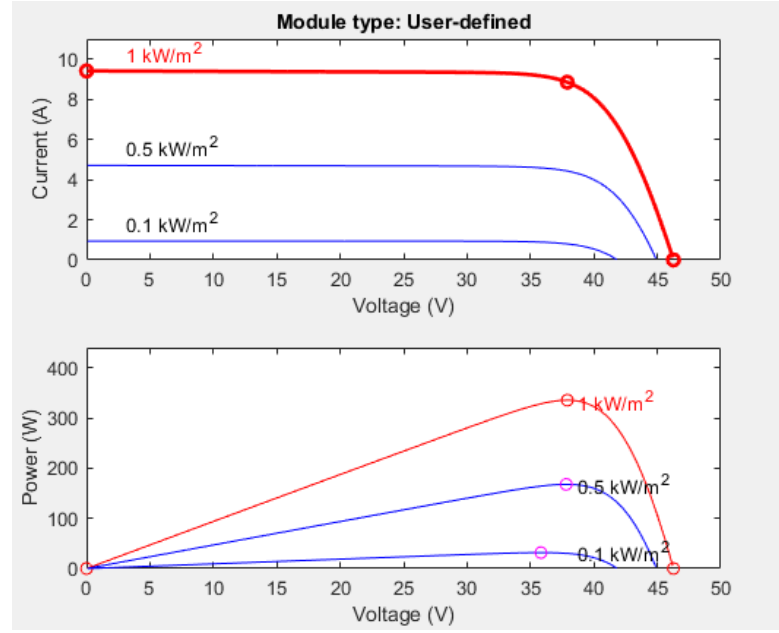


Figure 4.1 I–V and P–V characteristics of the PV module under STC

4.3. Field Observation Data

To test the model under varying operating conditions, two days with different weather conditions were selected as follows:

- **Day 1 (2 April 2026):** Clear sunny day with smoothly varying irradiance.
- **Day 2 (16 March 2026):** Partly cloudy day with inconsistent and rapidly varying irradiance.

Four quantities were recorded simultaneously at 10-minute intervals:

Table 4.2 Summary of Field Observations

Quantity	Day 1 (2 April 2026)	Day 2 (16 March 2026)
Number of readings	20	21
Time window	11:30 – 14:40	11:45 – 15:05
Weather condition	Clear sunny	Partly cloudy
Measured power range (W)	75 – 190	39 – 212
LDR irradiance range (W/m^2)	218 – 806	111 – 782
Solcast irradiance range (W/m^2)	605 – 1013	356 – 868
Ambient temperature range ($^{\circ}\text{C}$)	25 – 26	20 – 22

- Measured power output of the panel
- Irradiance from the calibrated LDR sensor (point-source measurement)
- Irradiance from Solcast satellite-derived data (regional estimate)
- Ambient temperature (obtained from on-site readings, assumed consistent for both point-source and regional estimates)

A total of 41 data points were collected across both days. The complete observation dataset is provided in Appendix B, while representative summary statistics are presented in Table 4.2.

The smooth progression of LDR irradiance on Day 1 confirms stable clear-sky conditions. In contrast, the variation observed in LDR-measured irradiance on Day 2 (with consecutive readings such as 252, 227, and 246 W/m^2 , rising sharply to 524 and 782 W/m^2 within 30 minutes) reflects the cloudy and transient nature of that day.

4.4. Simulation of Power Output

The observed values of irradiance and temperature were supplied as input to the verified Simulink model, and the steady-state power output was recorded. The simulation was run for both timestamps, one with the LDR-measured irradiance and another with the Solcast satellite-derived irradiance. All other parameters were kept constant during this simulation. The results were compared side by side based on the simulated power output.

Figure 4.2 presents the time-series comparison of measured power output, simulation output using LDR irradiance, and simulation output using Solcast irradiance. As shown, both days are included in the comparison.

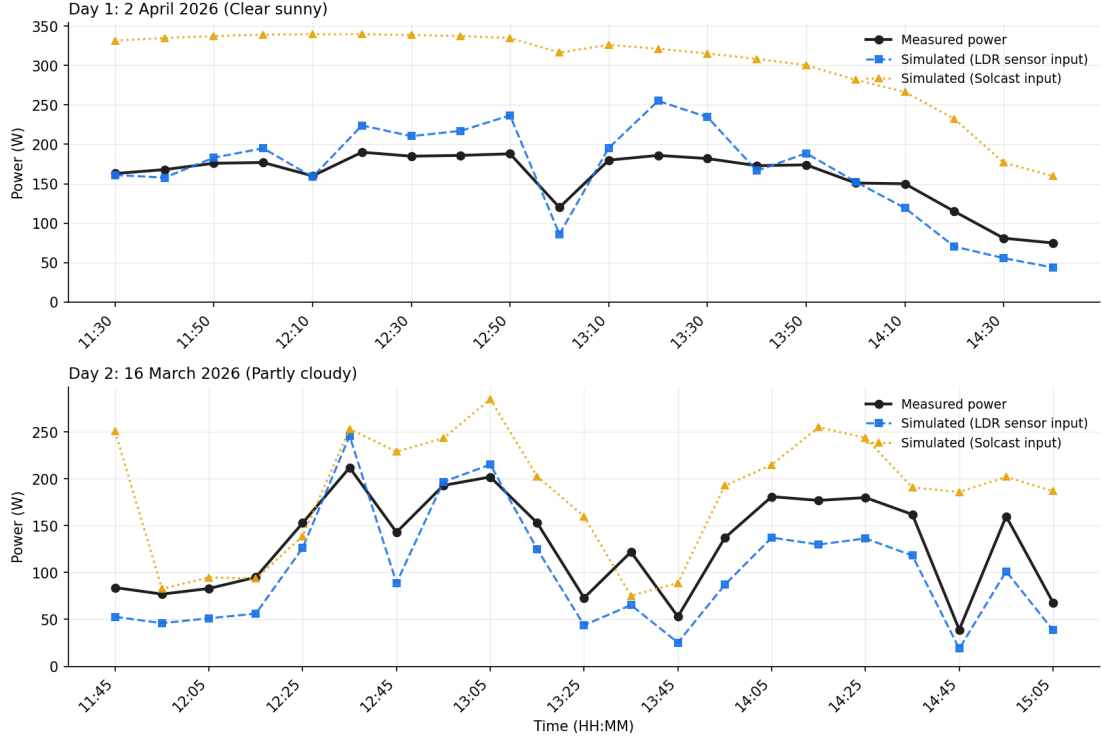


Figure 4.2 Time-series comparison of measured and simulated PV power output for (top) Day 1: 2 April 2026 (clear) and (bottom) Day 2: 16 March 2026 (partly cloudy).

Two observations are immediately evident from Figure 4.2:

- The LDR-irradiance simulation closely tracks the measured power values on both days, particularly during periods of stable irradiance (around solar noon on Day 1 and during the cleaner stretches on Day 2).
- The Solcast-irradiance simulation overpredicts power output, especially on Day 2. The simulated power remains high although the actual measured power dropped due to cloud cover. This is due to the use of spatial averaging in satellite-derived irradiance products, which most likely misses cloud cover in smaller observation areas.

4.5. Validation Metrics

Five standard error metrics were computed for each simulation method against the measured power. They were computed to find out the accuracy of the simulation, i.e., Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Pearson correlation coefficient (r), and coefficient of determination (R^2). Also, the mean bias was computed to identify the direction of error. Definitions of these metrics are provided in Section 3.7..

Table 4.3 Validation Metrics for Simulated vs Measured Power

Day	Method	RMSE (W)	MAE (W)	MAPE (%)	Bias (W)	r	R^2
Day 1 (2 April 2026)	LDR	31.10	25.15	17.22	+6.61	0.949	0.901
Day 1 (2 April 2026)	Solcast	145.28	142.86	92.92	+142.86	0.900	0.811
Day 2 (16 March 2026)	LDR	37.92	35.35	31.60	-30.52	0.935	0.874
Day 2 (16 March 2026)	Solcast	73.31	59.40	65.22	+53.47	0.640	0.409
Combined	LDR	34.76	30.38	24.58	-12.41	0.918	0.843
Combined	Solcast	114.23	100.11	78.73	+97.07	0.703	0.494

The results in Table 4.3 confirm that the LDR-irradiance simulation outperforms the Solcast-irradiance simulation across every metric. On the combined dataset, LDR-based simulation achieves $R^2 = 0.843$ with RMSE of 34.76 W, while Solcast-based simulation achieves only $R^2 = 0.494$ with RMSE of 114.23 W, which is about three times higher. Additionally, the Pearson correlation between measured and LDR-simulated power ($r = 0.918$) demonstrates a strong linear relationship. This indicates that the Simulink model is working correctly when supplied with realistic and instantaneous irradiance data.

The mean bias values are also informative. The Solcast-irradiance simulation has a bias of +97.07 W, meaning it overpredicts power by an average of 97 W. This is clear evidence of systematic overestimation due to satellite-data smoothing during cloudy conditions. The LDR-irradiance simulation has a much smaller bias of -12.41 W, with the slight underprediction caused by the LDR sensor's reduced sensitivity at low light levels (discussed further in Chapter 5).

The scatter plots in Figure 4.3 visualize the same relationship.

The points clustering tightly along the $x = y$ line in the case of the LDR sensor

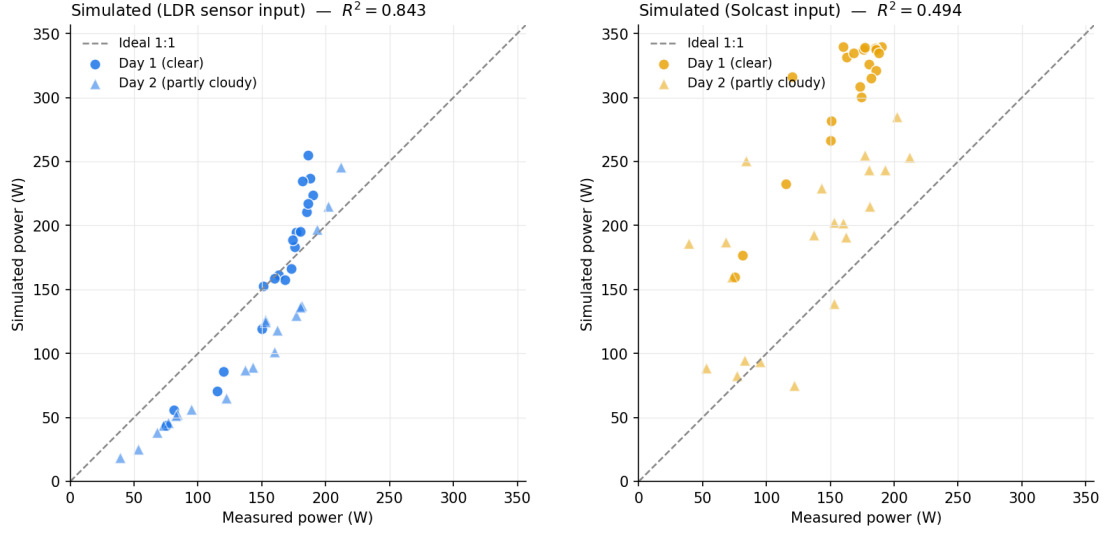


Figure 4.3 Scatter of simulated versus measured power. (Left) LDR-based simulation, $R^2 = 0.918$. (Right) Solcast-based simulation, $R^2 = 0.703$. The dashed line represents the ideal 1:1 relationship.

indicate accurate prediction, while the systematic upward deviation in the Solcast case confirms the overprediction.

4.6. Irradiance Band-wise Error Analysis

To investigate how simulation accuracy varies with operating conditions, the error pattern was examined across irradiance bands. The combined dataset was grouped into five bands of 200 W/m^2 width each, and within each band, the mean error percentage was computed. Results are summarised in Table 4.4 and Figure 4.4.

Table 4.4 Mean Simulation Error by Irradiance Band

Irradiance Band (W/m^2)	n	LDR Mean Error (%)	Solcast Mean Error (%)
Low (< 300)	10	+41.66	—
Med-Low (300–500)	6 (L) / 5 (S)	+37.40	-9.89
Med-High (500–700)	17 (L) / 10 (S)	+7.99	-101.02
High (700–900)	8 (L) / 11 (S)	-20.34	-64.49
Very High (> 900)	0 (L) / 15 (S)	—	-90.72

Error % defined as $(\text{Real} - \text{Simulated})/\text{Real} \times 100$; positive values indicate the simulation underpredicts measured power.

The LDR-based simulation underpredicts power at low irradiance (positive error of +37 to +42% in the $< 500 \text{ W/m}^2$ range), accurately predicts in the mid-

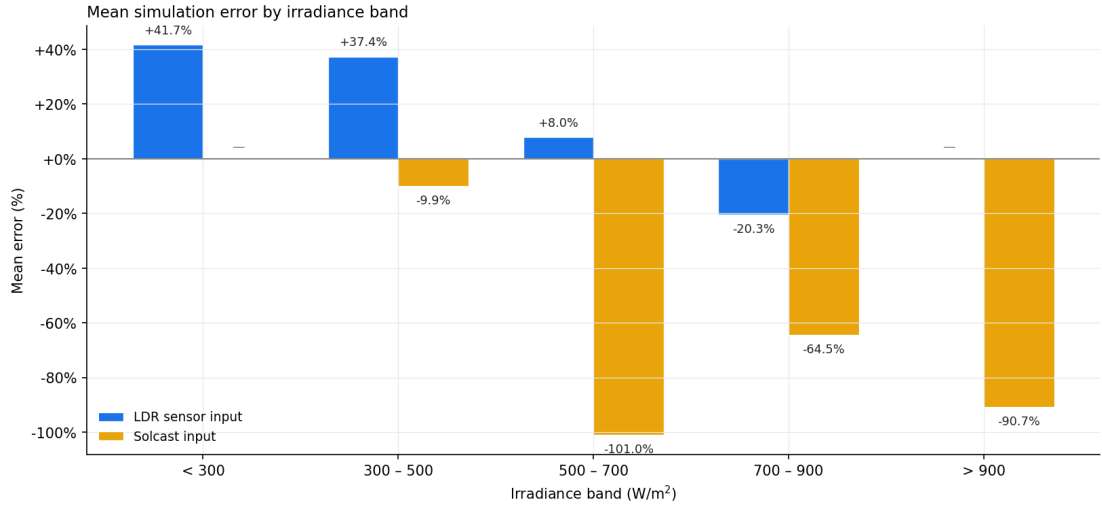


Figure 4.4 Mean simulation error percentage as a function of irradiance band, comparing LDR-irradiance and Solcast-irradiance simulation methods.

irradiance range (Med-High band, error of +8.0%), and slightly overpredicts at higher irradiance (−20.3% in the 700–900 W/m² band). This pattern is interpreted in detail in Chapter 5.

On the other hand, the Solcast-based simulation shows a monotonically increasing overprediction, with errors growing from −10% in the medium-low band to over −90% in the very-high band. This is due to the satellite irradiance estimates that frequently exceed 900 W/m² even during cloudy real-world conditions. This is reasonable as the data is calculated by the satellite using regional averaging.

Figure 4.5 plots the error as a function of irradiance for individual data points, illustrating the same pattern with finer resolution.

4.7. Comparison of Irradiance Reference Sources

The different power outputs obtained from the LDR-based and the Solcast-based irradiance motivate a closer look. Figure 4.6 directly compares the LDR sensor and Solcast satellite-derived irradiance readings for both observation days.

On Day 1 (clear conditions), both sources track the expected smooth solar profile. The Solcast source consistently reports values approximately 200–400 W/m² higher than the LDR source. On Day 2 (cloudy conditions), the difference widens

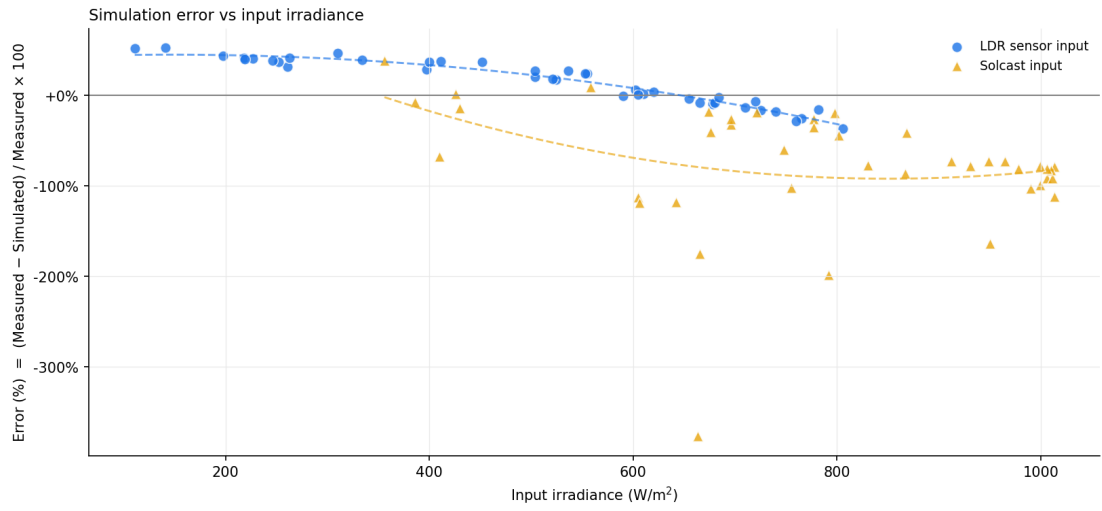


Figure 4.5 Simulation error percentage versus input irradiance for individual readings. The dashed lines indicate quadratic trend fits.

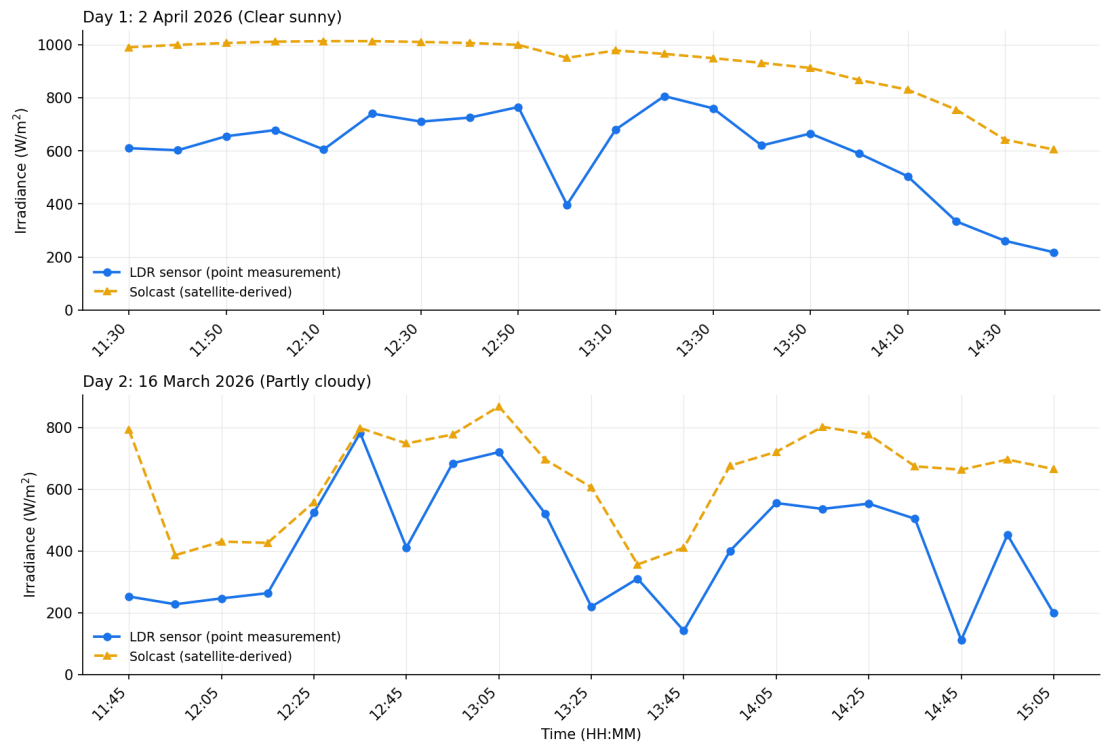


Figure 4.6 Comparison of LDR-based (point-source) and Solcast satellite-derived irradiance values for both observation days.

further as the Solcast source reports values exceeding 700 W/m^2 during some periods, but the LDR source reports values below 300 W/m^2 . This confirms that the spatial averaging used by Solcast to estimate irradiance fails to capture the presence of localised cloud cover at the observation site.

4.8. Daily Energy Comparison

For a practical context, the total energy generated during each observation window was computed. The power-time data was integrated using the trapezoidal rule. The time interval between successive samples was 10 minutes. The results are presented in Table 4.5 and Figure 4.7 as follows:

Table 4.5 Energy Generated During Each Observation Window

Date	Measured (Wh)	Sim – LDR (Wh)	Sim – Solcast (Wh)	LDR Error (%)	Solcast Error (%)
2 April 2026	530.0	552.0	1006.2	+4.15	+89.85
16 March 2026	457.8	351.0	645.0	-23.33	+40.89

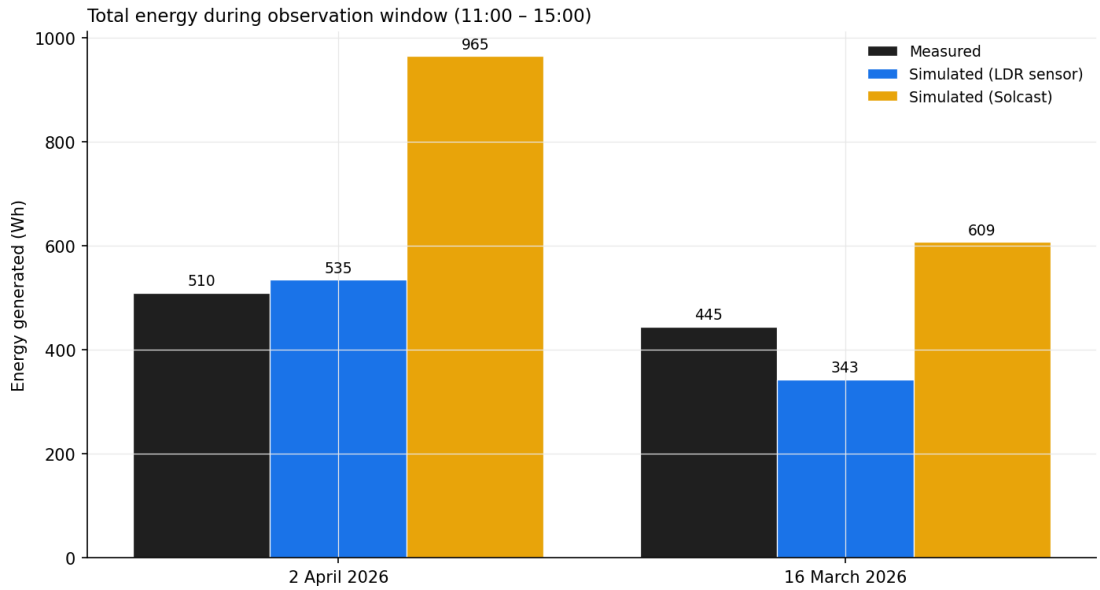


Figure 4.7 Energy generated during each observation window: measured versus simulated using LDR and Solcast irradiance.

For Day 1 (clear conditions), the LDR-irradiance simulation predicts energy generation within 4.2% of the measured value. On the other hand, the Solcast-irradiance simulation overpredicts the energy generation by 89.9%, which is almost double the actual generation. For Day 2 (cloudy conditions), the LDR-irradiance simulation

underpredicts the energy generation by 23.3%. The Solcast-irradiance simulation still overpredicts the generation amount by 40.9%. These results confirm that the Simulink model is capable of accurate energy generation prediction when supplied with irradiance measured locally at the site. The irradiance retrieved from satellite sources is unsuitable as a direct input for the site-specific PV simulation model.

4.9. Forecasting Performance

This section presents the results of the short-term forecasting discussed in Chapter 3. It is structured in two stages. The first is the SARIMA model used to forecast the next day’s hourly irradiance and temperature based on seven days of historical data. These forecasted values are then input into the validated Simulink digital twin to produce the hourly power forecast. The performance of each stage is evaluated separately and later compared with the others.

4.9.1. Dataset and Forecast Configuration

For training and validation of the SARIMA forecasting model, the hourly meteorological data for Kathmandu from July 2018 to June 2020 were used. The dataset contains hourly recorded global horizontal irradiance (GHI) and ambient air temperature. From this dataset, an eight-day continuous dataset (15–22 March 2020) was selected for forecasting analysis. This window was chosen for two major reasons. It is the same season as the project’s real-world observation, i.e., March–April. Also, the daytime data within this window are complete without any missing values.

The eight-day data were split into a training set with data from Days 1–7 and a test set with the data from Day 8. Thus, a total of 168 hourly observations for training and 24 hourly observations for testing were present. The SARIMA model was trained using the training set and forecast for Day 8. The forecast obtained was compared against the actual values. The accuracy metrics are calculated from the observations above. The model configuration used was $\text{SARIMA}(1, 1, 1) \times (1, 1, 1)_{24}$ with seasonal period $S = 24$ with full day’s hourly data.

4.9.2. Forecast Accuracy of Irradiance and Temperature

The trained SARIMA models were used to forecast irradiance and temperature for each of Day 8's hourly data. The standard error metrics were computed by comparing the forecasted values against the actual measured values. The results are summarised in Table 4.6.

Table 4.6 Forecast Accuracy Metrics for Day 8 (22 March 2020)

Metric	Irradiance	Temperature
RMSE	76.30 W/m ²	1.30 °C
MAE	34.69 W/m ²	1.06 °C
MAPE	26.03 %	6.56 %
R^2	0.921	0.964

The metrics show $R^2 = 0.921$ for irradiance and $R^2 = 0.964$ for temperature. The forecast observed is numerically closer to the actual values. Temperature is forecast more accurately than irradiance. The temperature is insensitive and stable throughout the day, whereas the irradiance is sensitive to instantaneous cloud cover. These factors for irradiance's inconsistency are not anticipated by the statistical models. The MAPE of 26% on irradiance is computed using only daytime values. The nighttime irradiance values are kept zero to avoid complexities. The graphs below (Figure 4.8) show the training data, the actual Day 8 values, and the SARIMA forecast for Day 8.

The model forecasts the variables accurately with very few problems. The most notable forecast value occurs around 12:00 on Day 8, where the actual irradiance dropped steeply from approximately 830 W/m² to 425 W/m² due to a transient cloud event. However, the SARIMA forecast smoothed throughout the time and predicted approximately 720 W/m². This represents one of the limitations of pure time-series forecasting. The model can accurately predict the expected daily pattern. The limitation is that it cannot anticipate localised cloud transients without sufficient weather data availability.

The SARIMA model's 95% confidence interval was calculated using the forecast variance. The floor was set to zero to prevent negative results. To test the relia-

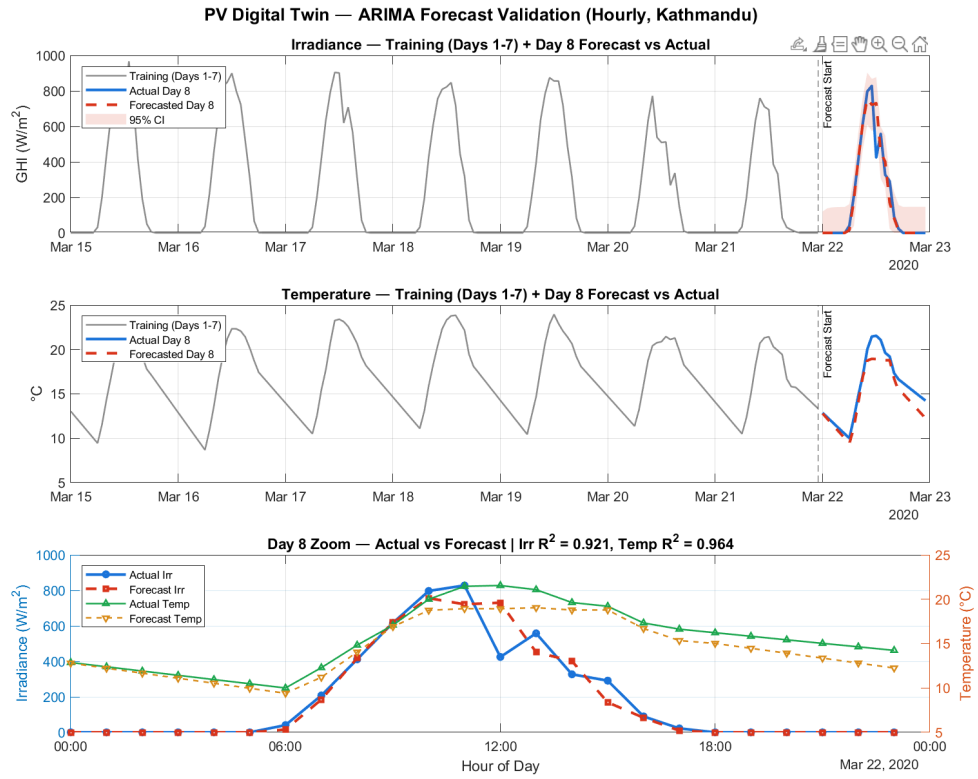


Figure 4.8 SARIMA forecast validation. (Top) Hourly irradiance over Days 1–7 and forecast vs actual on Day 8 with 95 % confidence interval shaded. (Middle) Hourly temperature with the same structure. (Bottom) Detailed comparison of actual and forecasted irradiance and temperature on Day 8.

bility of the model, the actual values were compared against the predicted values. On Day 8, 23 out of 24 hourly observations fell within the acceptable intervals. This result aligns closely with the theoretical 95% target. The model's uncertainty estimates are considered statistically accurate.

4.9.3. End-to-End Power Forecast

The digital twin forecasting pipeline forecasted irradiance and temperature for Day 8. These values were supplied as inputs to the validated Simulink model to produce the forecast of PV power output. The same Simulink model was also run with the actual measured irradiance and temperature inputs for Day 8, to produce a reference power profile. This comparison completely evaluates the forecast errors of the model. This way, the model's inaccuracies are isolated from the forecast's accuracy test. Since the real measured power output for 22 March 2020 was not available, this method of accuracy testing the SARIMA model is reasonable.

The hourly comparison is shown in Figure 4.9.

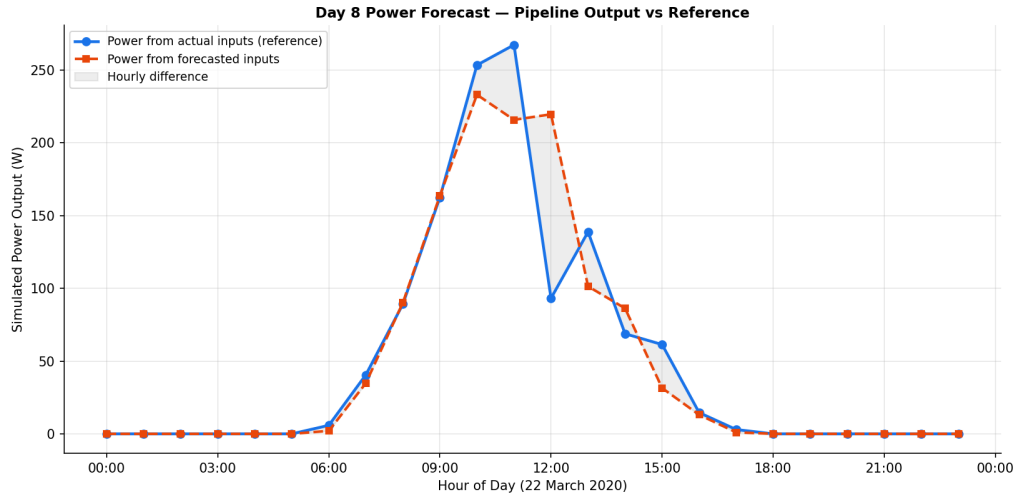


Figure 4.9 Day 8 PV power forecast. The reference power (blue) is the Simulink output when actual irradiance and temperature are supplied as inputs; the forecast power (orange dashed) is the Simulink output when SARIMA-forecasted inputs are supplied.

The end-to-end forecast accuracy is summarised in Table 4.7.

The forecast power profile correlates strongly with the reference profile ($R^2 =$

Table 4.7 End-to-End Forecast Pipeline Metrics for Day 8 Power Output

Metric	Value
RMSE	30.09 W
MAE	12.43 W
MAPE (daytime only)	32.29 %
Mean Bias (MBE)	−0.25 W
R^2	0.858
Forecast daily energy	1193 Wh
Reference daily energy	1199 Wh
Daily energy error	−0.50 %

0.858). The mean bias is -0.25 W, which indicates that the model forecasts the power accurately. The RMSE is attributed to the cloud transient around the early afternoon, as discussed in earlier sections.

The most engineering-relevant metric is daily energy yield. The forecast pipeline closely predicts an energy yield of 1193 Wh against a reference value of 1199 Wh. Despite the hourly fluctuations, the pipeline accurately predicts the total daily energy produced by the PV module.

4.10. Summary of Validation Results

This chapter has presented the experimental and simulation results in the following five parts: verification of the Simulink model at standard test conditions, two days of real-world observation data, simulation of power output using two irradiance reference sources, irradiance band-wise error analysis, and a short-term forecasting pipeline. The principal findings are summarized as follows:

1. The Simulink PV array model produced 335.1 W under STC (1000 W/m^2 , 25°C). Since the rating of the panel is 335 Wp, it confirms correct model parameterization.
2. When supplied with a site-specific irradiance value, the simulated power achieved $R^2 = 0.918$ with real power across both observation days. This validates the digital twin against field measurements.

3. Solcast satellite-derived irradiance gave lower simulation accuracy ($R^2 = 0.494$) because spatial averaging misses site-specific cloud transients.
4. Daily energy yield was predicted within 4.2% on a clear day using mobile irradiance. Thus, practical engineering accuracy can be obtained under favorable measurement conditions.
5. A clear error pattern was identified in the model's simulation. It under-predicted at low irradiance and slightly over-predicted at high irradiance. The phone sensor made errors in reading low-light irradiance. At high irradiance, the model neglected real-world losses. The losses are mainly due to panel heating, soiling, and cable losses.
6. The testing of the SARIMA forecasting model shows $R^2 = 0.921$ for irradiance and $R^2 = 0.964$ for temperature. It is statistically correct, as 95.8% of actual values fall within the predicted 95% confidence interval.
7. The complete digital twin forecasting pipeline is as follows: SARIMA forecast $\rightarrow$ Simulink digital twin $\rightarrow$ power output. This predicted the Day 8 daily energy generation with an error of only 0.5%. It shows that this framework is acceptable for short-term PV power forecasting.

These findings show that the developed digital twin framework represents the PV system under reference conditions. It accurately reproduces measured power when supplied with appropriate input data. It identifies the dominant sources of measurement and modeling error. It supports short-term forecasting through the integrated SARIMA forecasting pipeline. The interpretation of these findings alongside the sources of errors is presented in Chapter 5.

4.11. LDR Sensor Calibration Results

A Light-Dependent Resistor (LDR) was calibrated as a low-cost co-located irradiance sensor. The sensor was wired in a voltage-divider configuration with a $67\ \Omega$ fixed resistor, supplied by 5 V from an Arduino Uno. The midpoint voltage

was sampled by the Arduino’s analog input. It was recorded together with the simultaneous LDR irradiance reading.

A diffusing cover was placed over the LDR throughout the calibration to prevent sensor saturation at high irradiance. Since the LDR’s resistance drops sufficiently low to short the voltage divider, a bulb cover was used as a diffuser. The cover remained in place for both calibration and real-time monitoring to ensure consistency.

4.11.1. Calibration Procedure

A total of 21 paired voltage–irradiance readings were collected. The irradiance values ranged from 116 W/m² to 779 W/m². Readings were taken at different times of day to cover the full operating range.

A power-law model of the form

$$G = a \cdot V^b \quad \text{Equation 4.1}$$

was fitted to the data using nonlinear least-squares regression in MATLAB. Here, G is the reference irradiance in W/m² and V is the LDR voltage output in volts.

4.11.2. Calibration Results

The fitted equation obtained is:

$$G = 77.2 \cdot V^{3.663} \quad \text{Equation 4.2}$$

with a coefficient of determination $R^2 = 0.974$ between the fitted and reference irradiance. The calibration curve and measured data points are shown in Figure 4.10.

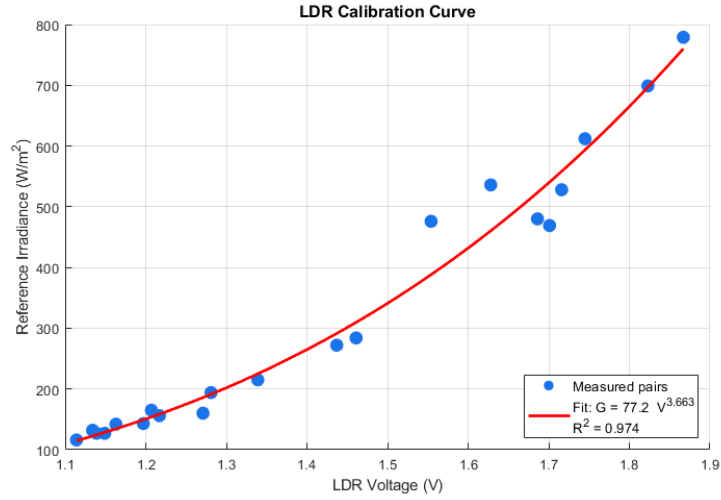


Figure 4.10 LDR voltage versus reference irradiance with fitted power-law calibration curve.

The calibration enables the LDR voltage output to be mapped to an estimated irradiance value. This completes the sensing element required for continuous site-specific measurement. However, the accuracy of this calibration is limited by the accuracy of the mobile application itself. The work presents the option of using a low-cost LDR-based irradiance sensing approach for the digital twin model.

4.12. Real-Time Operation Demonstration

To demonstrate the basics of a digital twin operating in real-time, the calibrated LDR sensor and a temperature sensor were connected to the Simulink model through a MATLAB interface. The Arduino read the LDR voltage and the temperature sensor output at one-second interval. The LDR voltage was converted to irradiance using the calibration equation 4.2, while the temperature sensor reading was used directly. Both values were sent to the serial communication. A MATLAB script received these values. Then the Simulink model was run with these live values of irradiance and temperature, and the corresponding power output was recorded.

4.12.1. System Configuration

The real-time pipeline consisted of:

- Arduino Uno reading the LDR voltage and the temperature sensor output
- LDR voltage converted to irradiance using the calibration equation
- Serial communication at 9600 baud over USB
- MATLAB script polling the serial port and updating the Simulink model
- Simulink model running with a 0.15 second simulation horizon per sample
- Continuous logging of timestamp, irradiance, temperature, and simulated power

4.12.2. Demonstration Results

A continuous run of 5 minutes was performed, generating 28 live samples. The recorded data is shown in Figure [4.11](#).

The simulated power followed the measured irradiance and temperature. This confirms the digital twin pipeline operates correctly in real-time with the LDR and temperature sensors as live inputs. The variations in irradiance produced corresponding variations in simulated power within the same sampling interval. This demonstration completes the transition of the digital twin from offline validation to live operation.

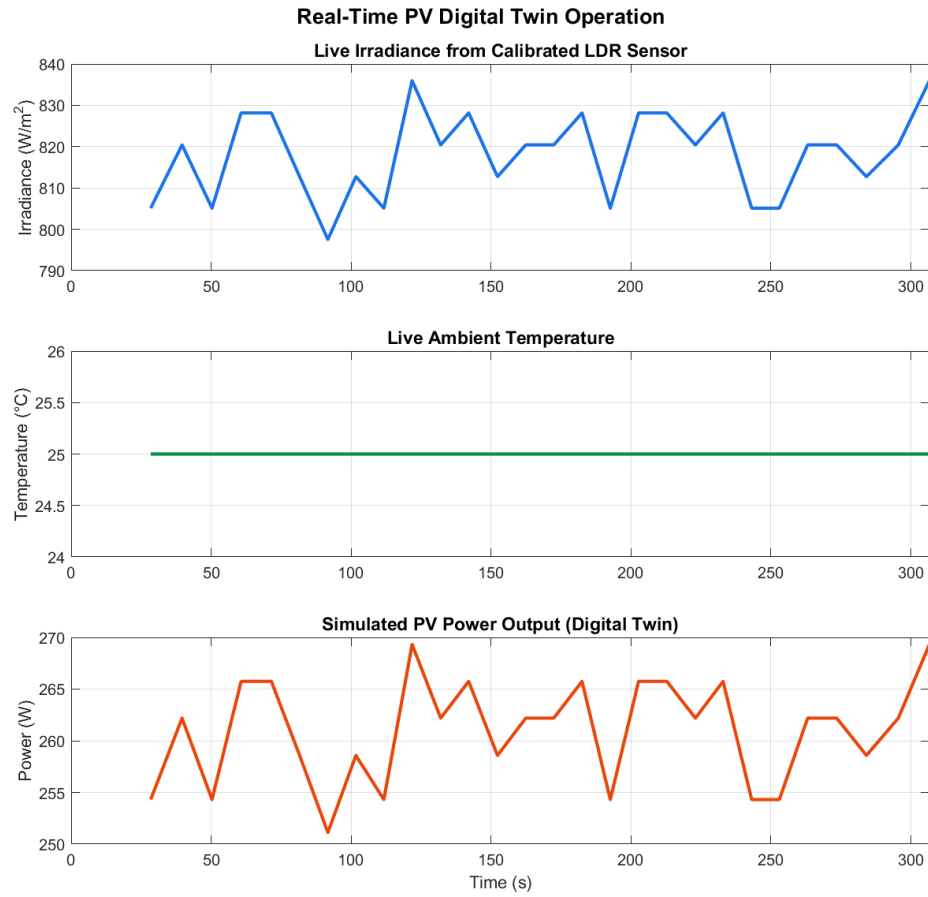


Figure 4.11 Real-time digital twin operation. (Top) Live irradiance measured by the calibrated LDR sensor. (Middle) Live ambient temperature from the temperature sensor. (Bottom) Simulated PV power output from the Simulink digital twin, computed in real-time from the live sensor inputs.

CHAPTER FIVE: DISCUSSION

This chapter examines the results presented in Chapter 4 and interprets their practical significance. It discusses how closely the developed digital twin represents the real PV system, compares the two irradiance measurement methods used in the study, and reflects on the implications of these findings for real-world PV monitoring at the site.

5.1. Interpretation of Validation Results

The Simulink-based digital twin developed in this study can be considered properly parameterised and a realistic representation of the actual PV module based on three main observations.

First, under Standard Test Conditions, the simulated power output is 335.1 W, which is very close to the manufacturer's rated value of 335 W, with a difference of only about 0.1%. Second, the Maximum Power Point obtained from the simulated P-V curve matches well with the datasheet value of $V_{mp} = 37.9$ V. This shows that the parameters used in the single-diode model are consistent with the module specifications. Third, when the model is supplied with irradiance and ambient temperature data obtained from the mobile app at the same location, the simulated power output shows a strong agreement with the measured power, with an R^2 value of 0.918 over the observation period.

Based on these observations, the Simulink model can be regarded as a valid digital twin for this study. The small differences between the simulated and measured results, discussed later in this chapter, are mainly due to limitations in the irradiance measurements rather than deficiencies in the developed model itself.

5.2. Comparison of Irradiance Reference Methods

This study clearly shows that the choice of irradiance data significantly affects the accuracy of PV system simulation. The LDR-based irradiance, although obtained using a simple and convenient consumer-level tool, performed well in this study. It closely matched the real system output with an R^2 value of 0.918, while the daily energy error remained low at about 4.2% on a clear day. In contrast, the Solcast satellite-based data showed lower performance, with an R^2 value of only 0.494 and errors exceeding 40% on both observed days.

The reason is quite intuitive. The mobile measurement is taken directly at the site, so it reflects the same sun and cloud conditions experienced by the PV panel. When clouds pass or partial shading occurs, the sensor captures these changes immediately, just as the PV system experiences them. However, satellite data operates on a much broader scale, averaging conditions over a large area of several square kilometres. Because of this, it cannot capture small and sudden local weather variations that strongly influence PV output.

Therefore, even though satellite data may appear more advanced, it is not always better for detailed site-specific analysis. It tends to smooth out real-world fluctuations, losing the fine details that are important for accurately reproducing system behaviour at a single location.

In simple terms, satellite-based irradiance is more suitable for the broader picture, such as estimating solar potential across a region or supporting long-term energy planning. However, for real-time simulation or for a PV digital twin of a specific system, local measurements are far more reliable. They may be simpler, but they remain closer to reality, and in engineering practice, that closeness often matters more than sophistication.

5.3. Structure of the Residual Errors

The residual errors in the Digital Twin model are not random. They represent the difference between the actual PV power and the predicted power at different

irradiance levels. The model anticipates lower power than actual measured power, with a positive error of about +40% at low irradiance levels, i.e., below 500 W/m². This is because real solar panels can perform relatively well under weak sunlight. This behavior is not fully captured by the model. Also, the model anticipates higher power than actual measured power, with a negative error of about 20% at high irradiance levels, i.e., between 700 and 900 W/m². This is because higher sunlight raises the panel temperature and reduces efficiency. This behavior is not completely captured in the model. But these opposite tendencies balance each other at the mid-range of irradiance levels, i.e., from 500 to 700 W/m². Here, the predicted and measured values come very close. Altogether, this reveals that the model has a consistent bias rather than purely random error. From this identified pattern, we can make improvements, such as including more representative data and better accounting for temperature and nonlinear behavior of the PV system.

5.3.1. Measurement-Side Error at Low Irradiance

The under-prediction at low irradiance is not mainly a problem of the Digital Twin model, but comes from the measurement system. In this project, the LDR sensor under-reports irradiance when the sunlight is weak. Because of this, the Simulink model gets a lower irradiance value and therefore predicts lower PV power. However, the real solar panel receives more actual sunlight than what is reported by the mobile sensor and produces higher power. This is also supported by the linear regression result, which shows a non-zero intercept of about +22 W at zero irradiance. This is physically not possible because a solar panel cannot produce power in complete darkness, which clearly shows measurement error. So, the apparent under-prediction at low irradiance is mainly an effect of inaccurate irradiance measurement. This is not a weakness of the model. This happens because mobile ambient-light sensors are not designed for scientific irradiance measurement and become less accurate under low-light conditions.

5.3.2. Model-Side Error at High Irradiance

The over-prediction at high irradiance is mainly due to limitations in the Digital Twin model. At irradiance levels above about 700 W/m^2 , the cell temperature increases significantly because of the NOCT effect, which reduces the efficiency of the panel. In addition, real systems experience small but continuous losses such as dust accumulation on the panel surface and resistance in cables, which are not included in the ideal Simulink model. Because of these missing effects, the model predicts higher power than what the actual PV system delivers. This is a common limitation of simplified PV models, as they work well in ideal conditions but cannot fully include all the small real-world losses in an actual solar system. This decomposition has direct engineering value: it separates errors that require better instrumentation from errors that require additional physical loss terms in the model, since the two are not the same problem and cannot be addressed by the same intervention.

5.3.3. Asymmetric Error Pattern Explanation

The residual error pattern here is not random. It actually follows a clear asymmetric trend across different irradiance levels. From the band-wise results in Chapter 4, the model tends to under-predict at low irradiance (around +40% error below 500 W/m^2), matches quite well in the mid-range, and then slightly over-predicts at higher irradiance (about -20% in the $700\text{--}900 \text{ W/m}^2$ range).

At low irradiance, the model underestimates power because some low-light behavior of the PV system is not fully captured. In reality, the system performs a bit better than what the simplified model expects.

In the mid-range, things balance out better. The operating conditions are closer to what the model assumes, so the predictions and measurements align quite closely.

At higher irradiance, the situation flips. The model starts to over-predict power mainly because cell temperature increases (NOCT effect), which reduces real efficiency. On top of that, practical losses like dust on the panel and cable resistance

also reduce actual output, but these are not included in the simulation. So the real system produces less than what the model assumes.

This asymmetric error pattern does not remain the same across the operating range but changes direction depending on the irradiance level. It happens because different physical effects become dominant at different operating levels, instead of one single source of error controlling everything.

Overall, the errors at the two ends come from different physical mechanisms acting in opposite ways. Interestingly, these effects partly cancel each other in the mid-range, which is why the model looks most accurate there.

5.4. Forecasting Pipeline Performance and Limitations

The SARIMA forecasting model achieved an R^2 of 0.921 for irradiance and 0.964 for temperature on the held-out validation day. On the same day, 95.8% of the actual observations fell within the model’s predicted 95% confidence interval. When these forecasts were passed through the digital twin to estimate power output, the overall end-to-end system produced a daily energy forecasting error of only 0.5%.

Overall, these results suggest that combining short-term statistical forecasting with a physics-based digital twin can produce fairly reliable day-ahead PV power predictions for engineering applications. The fact that 95.8% of actual values fall inside the predicted 95% confidence interval (which ideally should be 95%) is quite meaningful. It shows that the model’s uncertainty range is realistically tuned. In other words, the model is neither too confident nor too uncertain—the predicted range closely matches what actually happens in the system.

That said, there is still a clear limitation when it comes to sudden, short-lived weather changes, especially fast-moving cloud events. The biggest forecasting error on Day 8 occurred at 12:00, when irradiance dropped suddenly due to a passing cloud. The SARIMA model could not anticipate this because such a pattern was not present in the previous seven days of training data.

This highlights a known limitation of pure time-series forecasting methods. They

perform well for regular daily and seasonal trends, but struggle with random and sudden atmospheric changes unless extra external information is included. To handle this better, inputs like sky camera images or numerical weather prediction data would be needed. However, those hybrid approaches were not part of this study.

Another limitation is related to how the full pipeline was validated. Instead of directly comparing predicted power with actual measured output for the forecast day, the evaluation used a reference power profile generated by feeding actual inputs through the digital twin. This was necessary because the forecasting dataset (March 2020) does not match the actual measurement campaign period of this project (March–April 2026).

For future work, it would be better to test the same forecasting pipeline on a real multi-day dataset collected directly from the installation site. This would allow a direct comparison between predicted and measured power on a daily basis. It would also give a more practical and realistic understanding of how the model behaves in actual operating conditions, rather than relying on a reference-based evaluation.

5.5. Practical Implications for PV Monitoring at Pulchowk

The results of this work have clear practical value for operating and monitoring the PV system at the Department of Electrical Engineering, Pulchowk Campus.

The digital twin we developed can be used as a simple reference to check how our real PV system is performing. When we give it proper local irradiance data from the same location, it usually matches the actual power output quite well. But if we start seeing a clear and repeated difference between the predicted and measured power, it usually means something is wrong in the system. This could be due to dust on the panels, partial shading from nearby objects, or normal aging of the system over time. These problems are not always easy to notice just by looking at raw power data, but the model makes it easier to understand that something

unusual is happening.

The forecasting result is also quite useful in practice. With around 0.5% error in daily energy prediction, it is accurate enough for basic planning work. It can help in deciding how to manage loads in the department, when to charge batteries, and when extra energy can be sent to the grid. So it is not just a simulation model for study purposes, but something that can actually help in daily operation and planning of the PV system.

At the same time, the study highlights an important limitation in using satellite-derived irradiance for precise, real-time simulation. The results show that it is not reliable enough at the local scale of a PV installation like Pulchowk. This naturally leads to the need for a co-located irradiance measurement system installed directly at the site. The LDR-based sensing approach explored in this work offers a low-cost starting point for such measurement. However, since its calibration depends on a mobile application reference, and that reference itself is not fully accurate, the resulting measurements still contain uncertainty. Because of this, the calibration method needs further refinement before the LDR-based system can be confidently used as the main irradiance input for the digital twin in real operational conditions.

CHAPTER SIX: CONCLUSION AND FUTURE WORK

6.1. Summary of Key Works

This project is focused on developing a digital twin of a Solar PV system that is present in the Renewable Lab at the Electrical Engineering department. The twin was built in Simulink using the single diode equivalent circuit with the P&O algorithms to ensure operation of system in MPP, boost converter to boost up the voltage level to 65V, single phase grid tied inverter and the associated control system as a digital copy of the existing system. To validate the developed model, it was checked against physical system by manually entering value of irradiance and temperature value at the same instance.

A weather monitoring station is installed on the rooftop of the Department of Electronics and Computer Engineering. Initially, the project aimed to provide real-time solar irradiance and temperature data directly from the station to the Simulink model. However, this could not be implemented due to issues with the data logger system. Despite this limitation, historical weather data from the station was successfully obtained and utilized for the development of the SARIMA forecasting model. The collected dataset is used as the basis for forecasting solar irradiance for short-term photovoltaic power prediction. For real-time integration of irradiance input to simulink model, a low cost pyranometer was developed using LDR, arduino, resistor. The developed pyranometer was calibrated using the Irradiance meter app in mobile. It couldn't be calibrated using real pyranometer because of higher cost and resource limit which remained a limitation in the study. The model used low cost self made pyranometer as an input for irradiance value to simulate the real-time data. At same instance, the power of physical PV system of renewable lab was observed in order to validate developed digital twin. The data were collected at every 10 minutes taken over 2 days during peak sun hour. The selected days were partly clouded day and a clear sunny day.

The SARIMA model was used to forecast solar irradiance for a selected test day. These forecasted irradiance values were then fed into the simulation model to predict the PV system energy output. The predicted power output was compared with the actual system power output. The forecasting model was validated based on the observed errors presented in the results section.

6.2. Conclusions

When the developed digital twin model was simulated under Standard Test Conditions, it produces output as 335.1 Watts. It was nearly close to the 335 Watts which is the panel output at STC mentioned in nameplate. Also, the model behaved properly following the real power measurements with an R^2 value of 0.918. When the model was fed with the sunlight data from the same spot as the panel, the results confirmed that developed digital twin model was set up correctly and truly showed how the installed solar module behaved.

A low-cost pyranometer was successfully developed using an LDR, Arduino Uno, and a fixed resistor ($67\ \Omega$) configured in a voltage-divider arrangement. The developed sensor was calibrated against mobile application (Irradiance meter) readings using a relationship in the form $G = a \cdot V^b$, where the calibration coefficients were determined through non linear least square regression in MATLAB. The calibrated LDR output was interfaced with the MATLAB through serial communication, enabling the measured irradiance values to be passed as dynamic inputs to the Simulink Digital Twin model in real-time. The real-time demonstration confirmed that the complete sensing pipeline from live LDR voltage reading to irradiance conversion to Simulink power output operated correctly within the same sampling interval, with simulated power responding continuously to changes in live irradiance and temperature conditions. However, it is acknowledged that the calibration accuracy of the developed pyranometer is inherently limited by the accuracy of the mobile application sensor used as the reference, as smartphone ambient light sensors are not designed for scientific irradiance measurement. Despite this limitation, the objective of developing and integrating a low cost hardware

irradiance sensing layer with the Simulink Digital Twin was successfully achieved, with the calibrated LDR sensor enabling field-validation accuracy of $R^2 = 0.918$. The comparative analysis was performed on measured, simulated and forecasted power output for the observation days taken on both clear and cloudy day. The Digital Twin model with an irradiance input from developed pyranometer showed robust agreement with the physical system's measured output yielding an R^2 of 0.918 and a Root Mean Square Error (RMSE) of 34.76 W. Its daily energy prediction on the clear day maintained an accuracy within 4.15 %. Also, irradiance data derived from the internet based Solcast platform exhibited poorer performance with an R^2 of 0.494 and a mean bias of +97.07 W which confirms inappropriate for site specific simulations. The SARIMA forecasting model achieved an R^2 of 0.858 coupled with a daily energy error of merely 0.5% indicating its reliable application for short term power prediction. A systematic deviation was identified between the Digital Twin's output and the physical system's actual output. This deviation clearly manifest underprediction at low irradiance and overprediction at high irradiance provided valuable insights in designing more robust system. The losses which include phenomena such as panel heating, soiling, and cable resistance are not incorporated into the current simulation model. Such understanding can be included in the system by using other Machine learning techniques.

6.3. Recommendations for Future Work

This project successfully achieved its major objectives. However, certain limitations and practical challenges identified during the study provide opportunities for future improvements and further development.

The most important concern is calibration of developed LDR based pyranometer. In this study, the LDR was calibrated against a mobile application (Irradiance meter) which itself carries some measurement errors. In future work, it should be calibrated with a certified reference pyranometer to eliminate calibration error and improve the accuracy of the real-time irradiance measurement. Alternatively, a real pyranometer can be employed which can act as a weather station.

The SARIMA forecasting model employed in this study is a linear statistical method and is limited in its ability to capture sudden nonlinear atmospheric disturbances such as instantly varying cloud which were identified as the primary source of hourly forecast error in the result. More advanced forecasting techniques such as Long Short-Term Memoery (LSTM) or deep learning methods can be employed in future which suits better for non linear temporal change in solar irradiance.

The current Digital Twin framework operates in an open loop manner where sensor inputs are fed into the Simulink model without any feedback mechanism from the physical system. A closed loop real time Digital Twin should be developed in future where real time measurements from the physical PV system continuously update the model state and allows automatic recalibration when deviations between simulated and measured output exceed a defined threshold. This would optimize the forecasting model towards accuracy.

In future, the model can be added with a potential for developing anomaly detection and fault identification by comparing the deviations. If the deviation between the Digital Twin output and the measured power consistently exceeds a defined threshold at same instance of operation, it indicates module degradation, component failure or external failures indicating the need predictive maintenance.

Finally, the current framework is developed and validated for a small module in the laboratory. In future, the scale of the framework can be increased to design the Digital Twin architecture for larger PV installations.

REFERENCES

- Alao, R. O., Zhang, C., & Muyeen, S. M. (2024). A review on digital twin application in photovoltaic energy systems: Challenges and opportunities. *JMST Advances*. <https://doi.org/10.1007/s44204-024-00093-1>
- Antonanzas, J., Osorio, N., Escobar, R., Urraca, R., Martinez-de-Pison, F., & Antonanzas-Torres, F. (2016). Review of photovoltaic power forecasting. *Solar Energy*, 136, 78–111. <https://doi.org/10.1016/j.solener.2016.06.069>
- Atallah, A., Abdelaziz, A., & Jumaah, R. (2014). Implementation of perturb and observe mppt of pv system with direct control method using buck and buck-boost converters. *Emerging Trends in Electrical, Electronics & Instrumentation Engineering (EEIEJ)*, 1, 31–44.
- Dhimish, M., Holmes, V., Mehrdadi, B., & Dales, M. (2018). Comparing mamdani sugeno fuzzy logic and rbf ann network for pv fault detection. *Renewable Energy*, 117, 257–274.
- Fong Mollineda, L., & García, J. (2024). Digital twin of a photovoltaic system. *ITEGAM-JETIA*, 10.
- Gaviria, J. F., Narvaez, G., Guillen, C., Giraldo, L., & Bressan, M. (2022). Machine learning in photovoltaic systems: A review. *Renewable Energy*, 196. <https://doi.org/10.1016/j.renene.2022.06.105>
- International Energy Agency. (2023, October). *World Energy Outlook 2023* (tech. rep.). IEA. Paris, France. <https://www.iea.org/reports/world-energy-outlook-2023>
- International Renewable Energy Agency (IRENA). (2023). *Renewable Power Generation Costs in 2022* (tech. rep.). IRENA. Abu Dhabi, UAE. <https://www.irena.org/Publications/2023/Aug/Renewable-Power-Generation-Costs-in-2022>
- Karunanidhi, B., et al. (2024). Development of a digital twin framework for a pv system to resolve partial shading. *Mathematical Problems in Engineering*, 2024, 1–15. <https://doi.org/10.1155/2024/8374487>

- Kolahi, M., et al. (2024). Digital-pv: A digital twin-based platform for autonomous aerial monitoring of large-scale photovoltaic power plants. *Energy Conversion and Management*. <https://doi.org/10.1016/j.enconman.2024.118963>
- Massel, L., Shchukin, N., & Cybikov, A. (2021). Digital twin development of a solar power plant. *E3S Web of Conferences*, 289, 03002.
- Okeke, G., et al. (2025). An evaluation of renewable energy technologies and their applications in electricity generation. *Energy Research Journal*.
- Quan, H., Srinivasan, D., & Khosravi, A. (2014). Short-term load and wind power forecasting using neural network-based prediction intervals. *IEEE Transactions on Neural Networks and Learning Systems*, 25(2), 303–315. <https://doi.org/10.1109/TNNLS.2013.2276053>
- Sharma, S., & Mishra, S. (2023). Electricity demand estimation using arima forecasting model. <https://doi.org/10.3233/ATDE221331>
- Solar Power Europe. (2023). *Global Market Outlook for Solar Power 2023–2027* (tech. rep.). Solar Power Europe. Brussels, Belgium.
- Solcast. (2020). Solar Irradiance and PV Power Data: Live and Forecast API for Kathmandu, Nepal [Satellite-derived irradiance data accessed for geographic coordinates 27.6588°N, 85.3247°E].
- Stein, J. (2012). *The photovoltaic performance modeling collaborative (pvpmc)* (tech. rep.). Sandia National Laboratories.
- Yu, W., Liu, G., Zhu, L., & Yu, W. (2025). Digital twin: A literature review of concepts, technologies, and applications. *IEEE Access*, 13, 183075–183095. <https://doi.org/10.1109/ACCESS.2025.3624583>
- Zhang, N., et al. (2025). Physics-based digital twin modeling of pv modules without historical data. *Energy Reports*. <https://doi.org/10.1016/j.egyr.2025.11.039>

APPENDICES

Appendix A: Raw Observation Data

A.1 Day 1 – 2 April 2026 (Clear Sunny)

Time	P_{meas} (W)	G_{LDR} (W/m ²)	$G_{Solcast}$ (W/m ²)	T (°C)	$P_{sim,LDR}$ (W)	$P_{sim,Solcast}$ (W)
11:30	163	610	990	25	161.20	331.53
11:40	168	602	999	25	157.80	334.78
11:50	176	655	1006	25	183.10	337.30
12:00	177	678	1011	25	195.00	339.09
12:10	160	605	1013	25	158.70	339.57
12:20	190	740	1013	25	224.00	339.57
12:30	185	710	1010	25	210.40	338.61
12:40	186	725	1006	25	217.10	337.30
12:50	188	765	999	25	236.70	334.78
13:00	120	397	950	25	86.06	316.40
13:10	180	680	978	26	195.40	326.07
13:20	186	806	965	26	255.10	321.17
13:30	182	760	949	26	234.50	315.27
13:40	173	620	931	26	166.50	308.29
13:50	174	665	912	26	188.70	300.39
14:00	151	590	867	26	152.60	281.83
14:10	150	504	830	26	119.20	266.29
14:20	115	334	755	26	70.45	232.29
14:30	81	261	642	26	55.78	176.84
14:40	75	218	605	26	43.88	159.79

Table 7.1 Day 1 raw observations (2 April 2026, clear sunny).

A.2 Day 2 – 16 March 2026 (Partly Cloudy)

Time	P_{meas} (W)	G_{LDR} (W/m ²)	$G_{Solcast}$ (W/m ²)	T (°C)	$P_{sim,LDR}$ (W)	$P_{sim,Solcast}$ (W)
11:45	84	252	792	21	52.82	250.58
11:55	77	227	386	21	46.00	82.63
12:05	83	246	430	21	51.26	94.82
12:15	95	263	426	22	56.12	93.55
12:25	153	524	558	22	126.40	138.90
12:35	212	782	798	22	245.80	253.32
12:45	143	411	748	22	89.01	228.95
12:55	193	684	777	22	196.80	243.42
13:05	202	720	868	22	215.20	284.92
13:15	153	521	696	21	125.00	202.51
13:25	73	219	606	21	43.83	159.98
13:35	122	310	356	21	65.34	75.23
13:45	53	141	410	21	25.26	88.92
13:55	137	400	676	21	87.02	192.66
14:05	181	555	721	21	137.30	214.93
14:15	177	536	802	21	129.90	255.05
14:25	180	553	777	20	136.40	243.64
14:35	162	504	674	20	118.10	190.82
14:45	39	111	663	20	18.87	185.98
14:55	160	452	696	20	101.30	202.06
15:05	68	198	665	20	38.44	186.96

Table 7.2 Day 2 raw observations (16 March 2026, partly cloudy).

A.3 LDR Calibration Dataset

LDR Voltage (V)	Irradiance (W/m ²)
1.867	779
1.823	699
1.437	272
1.339	215
1.281	194
1.271	160
1.217	156
1.197	143
1.163	142
1.149	127
1.139	127
1.114	116
1.134	132
1.716	528
1.686	480
1.628	536
1.554	476
1.745	612
1.207	165
1.461	284
1.701	469

Table 7.3 LDR calibration measurements used for irradiance estimation.

Appendix B: MATLAB Batch Simulation Script

The MATLAB script below was used to drive the Simulink model with the Field-measured irradiance and temperature data are used as inputs to the simulation. The resulting CSV is used for validation in Chapter 4.

```
1  %% Batch PV Simulation
2  clc; clear;
3
4  % Configuration
5  INPUT_CSV    = 'input_data.csv';
6  OUTPUT_CSV   = 'observation_data_with_sim.csv';
7  USE_COLUMN   = 'Irradiance';
8  SIM_TIME     = '2';
9  MODEL_NAME   = 'Grid_connected_single_phase_solar';
10
11 % Block paths
12 IRR_BLOCK     = [MODEL_NAME '/Constant'];
13 TEMP_BLOCK    = [MODEL_NAME '/Constant1'];
14
15 % Load data
16 T = readtable(INPUT_CSV);
17 fprintf('Loaded %d rows from %s\n', height(T), INPUT_CSV);
18
19 irr_vec = T.(USE_COLUMN);
20 temp_vec = T.Temperature;
21
22 % Open Simulink model
23 load_system(MODEL_NAME);
24
25 % Loop through each observation
26 n = height(T);
27 sim_power = zeros(n, 1);
28
29 for i = 1:n
30     G = irr_vec(i);
31     Tc = temp_vec(i);
```

```

32
33     if G < 1
34         sim_power(i) = 0;
35         continue;
36     end
37
38     set_param(IRR_BLOCK, 'Value', num2str(G));
39     set_param(TEMP_BLOCK, 'Value', num2str(Tc));
40
41     simOut = sim(MODEL_NAME, 'StopTime', SIM_TIME);
42
43     P = simOut.power_out;
44     if isobject(P), P = P.Data; end
45
46     % Steady-state mean (last 10% of samples)
47     n_steady = max(1, round(length(P) * 0.9));
48     sim_power(i) = mean(P(n_steady:end));
49
50     fprintf('Row %2d: G=%4.0f W/m^2, T=%2.0f C -> P = %6.2f W\n', ...
51           i, G, Tc, sim_power(i));
52 end
53
54 % Save results
55 T.Simulated_Power_W = round(sim_power, 2);
56 writetable(T, OUTPUT_CSV);
57
58 % Comparison metrics if Power_Real present
59 if any(strcmp(T.Properties.VariableNames, 'Power_Real'))
60     err = T.Power_Real - T.Simulated_Power_W;
61     fprintf('\nMean error : %.2f W\n', mean(err));
62     fprintf('RMSE       : %.2f W\n', sqrt(mean(err.^2)));
63     fprintf('MAE        : %.2f W\n', mean(abs(err)));
64 end

```

Listing 7.1 Batch PV simulation script.

Appendix C: Python SARIMA Forecasting Script

The script below trains the SARIMA models for hourly irradiance and temperature on Days 1–7, produces a 24-hour forecast for Day 8, and reports the validation metrics quoted in Section 4.9.

```
1
2 clc; clear; close all;
3
4 %% -- SECTION 1 : Load Historical 8-Day Window
   -----
5 fprintf('=== STEP 1: Loading 8-day historical window ===\n');
6
7 CSV_FILE = 'kathmandu_8day_window.csv';
8
9 opts = detectImportOptions(CSV_FILE);
10 opts = setvaropts(opts, 'Timestamp', 'InputFormat', 'yyyy-MM-dd
    HH:mm:ss');
11 rawTbl = readtable(CSV_FILE, opts);
12 data = table2timetable(rawTbl);
13
14 % Confirm we have 8 days of hourly data = 192 rows
15 fprintf('Loaded %d rows from %s to %s\n', height(data), ...
16     char(data.Timestamp(1)), char(data.Timestamp(end));
17
18 % Sampling interval (hourly data)
19 interval_min = 60; % 60 minutes between samples
20 S = 24; % 24 hourly samples per day
21 horizon = 24; % forecast 24 hours = 1 day
22
23 %% -- SECTION 2 : Split into Training (7 days) and Test (1 day)
   --
24 fprintf('\n=== STEP 2: Splitting data ===\n');
25
26 n_train = 7 * S; % 168 rows = days 1-7
27 trainData = data(1:n_train, :);
28 testData = data(n_train+1:end, :);
```

```

29
30 fprintf('Training set: %d rows (%s to %s)\n', height(trainData)
    , ...
31     char(trainData.Timestamp(1)), char(trainData.Timestamp(end)
    ));
32 fprintf('Test set      : %d rows (%s to %s)\n', height(testData),
    ...
33     char(testData.Timestamp(1)), char(testData.Timestamp(end)))
    ;
34
35 %% -- SECTION 3 : Stability Fix (nighttime zeros)
    -----
36 fprintf('\n=== STEP 3: Stability fix ===\n');
37
38 rng(42); % reproducibility
39 trainData.Irradiance = trainData.Irradiance + 1e-6 * randn(
    height(trainData), 1);
40 trainData.Temperature = trainData.Temperature + 1e-6 * randn(
    height(trainData), 1);
41
42 trainData.Irradiance(trainData.Irradiance < 0) = 0;
43
44 %% -- SECTION 4 : Define & Train SARIMA Model
    -----
45 fprintf('\n=== STEP 4: Defining SARIMA(1,1,1)(1,1,1)[24] model
    ===\n');
46
47 % Hourly data -> S=24 (24 samples per day)
48 % AR(1) + I(1) + MA(1) at non-seasonal level
49 % Seasonal AR(1) + I(1) + MA(1) at lag-24
50 Mdl = arima('ARLags',      1, ...
51             'D',           1, ...
52             'MALags',      1, ...
53             'Constant',    0, ...
54             'Seasonality', S, ...
55             'SARLags',     S, ...

```

```

56         'SMALags',      S);
57
58 fprintf('\n=== STEP 5: Training models on Days 1-7 (1-3 minutes
        ) ==\n');
59
60 fprintf('  Training Irradiance model...\n');
61 estIrr = estimate(Mdl, trainData.Irradiance, 'Display','off');
62 fprintf('  Done.\n');
63
64 fprintf('  Training Temperature model...\n');
65 estTemp = estimate(Mdl, trainData.Temperature, 'Display','off')
        ;
66 fprintf('  Done.\n');
67
68 %% -- SECTION 6 : Forecast Day 8
        -----
69 fprintf('\n=== STEP 6: Forecasting Day 8 ==\n');
70
71 [irrF, irrMSE] = forecast(estIrr, horizon, 'Y0', trainData.
        Irradiance);
72 [tempF, tempMSE] = forecast(estTemp, horizon, 'Y0', trainData.
        Temperature);
73
74 % Physical constraint
75 irrF(irrF < 0) = 0;
76
77 % 95 % confidence intervals
78 irrCI_upper = irrF + 1.96 * sqrt(irrMSE);
79 irrCI_lower = max(irrF - 1.96 * sqrt(irrMSE), 0);
80
81 % Forecast timestamps
82 tForecast = trainData.Timestamp(end) + hours(1:horizon);
83 tForecast = tForecast(:);
84
85 fprintf('Forecast range: %s to %s\n', ...
86         char(tForecast(1)), char(tForecast(end)));

```

```

87 fprintf('Peak forecasted irradiance : %.1f W/m^2\n', max(irrF))
    ;
88 fprintf('Peak actual      irradiance : %.1f W/m^2\n', max(
    testData.Irradiance));
89
90 %% -- SECTION 7 : Compute Validation Metrics
    -----
91 fprintf('\n=== STEP 7: Forecast Accuracy Metrics ===\n');
92
93 actualIrr = testData.Irradiance;
94 actualTemp = testData.Temperature;
95
96 % Irradiance metrics
97 err_i = irrF - actualIrr;
98 rmse_i = sqrt(mean(err_i.^2));
99 mae_i = mean(abs(err_i));
100 % Avoid div-by-zero for nighttime (use only daytime, irr > 5)
101 day_mask = actualIrr > 5;
102 mape_i = mean(abs(err_i(day_mask) ./ actualIrr(day_mask))) *
    100;
103 r_i = corrcoef(actualIrr, irrF); r_i = r_i(1,2);
104 R2_i = r_i^2;
105
106 % Temperature metrics
107 err_t = tempF - actualTemp;
108 rmse_t = sqrt(mean(err_t.^2));
109 mae_t = mean(abs(err_t));
110 mape_t = mean(abs(err_t ./ actualTemp)) * 100;
111 r_t = corrcoef(actualTemp, tempF); r_t = r_t(1,2);
112 R2_t = r_t^2;
113
114 fprintf('\n+-----+\n');
115 fprintf('| IRRADIANCE FORECAST METRICS | \n');
116 fprintf('+-----+\n');
117 fprintf('| RMSE : %7.2f W/m^2 | \n',
    rmse_i);

```



```

118 fprintf('|   MAE           : %7.2f W/m^2           |\n',
        mae_i);
119 fprintf('|   MAPE (day) : %7.2f %%           |\n',
        mape_i);
120 fprintf('|   R^2           : %7.4f           |\n',
        R2_i);
121 fprintf('+-----+\n');
122
123 fprintf('\n+-----+\n');
124 fprintf('|   TEMPERATURE FORECAST METRICS           |\n');
125 fprintf('+-----+\n');
126 fprintf('|   RMSE           : %7.2f degC           |\n',
        rmse_t);
127 fprintf('|   MAE           : %7.2f degC           |\n',
        mae_t);
128 fprintf('|   MAPE           : %7.2f %%           |\n',
        mape_t);
129 fprintf('|   R^2           : %7.4f           |\n',
        R2_t);
130 fprintf('+-----+\n');
131
132 %% -- SECTION 8 : Save Results CSV
        -----
133 fprintf('\n=== STEP 8: Saving results ===\n');
134
135 resultsTbl = table(tForecast, ...
136                   actualIrr, irrF, irrCI_lower, irrCI_upper,
137                   ...
138                   actualTemp, tempF, ...
139                   'VariableNames', { ...
140                       'Timestamp', ...
141                       'Actual_Irradiance', '
Forecast_Irradiance', ...
142                       'CI_Lower', 'CI_Upper', ...
143                       'Actual_Temperature', '
Forecast_Temperature'}});

```

```

143
144 writetable(resultsTbl, 'forecast_vs_actual_day8.csv');
145 fprintf('Saved: forecast_vs_actual_day8.csv\n');
146
147 % Save metrics table
148 metricsTbl = table( ...
149     {'RMSE'; 'MAE'; 'MAPE'; 'R^2'}, ...
150     [rmse_i; mae_i; mape_i; R2_i], ...
151     [rmse_t; mae_t; mape_t; R2_t], ...
152     'VariableNames', {'Metric', 'Irradiance', 'Temperature'});
153 writetable(metricsTbl, 'forecast_metrics.csv');
154 fprintf('Saved: forecast_metrics.csv\n');
155
156 %% -- SECTION 9 : Plots
157     -----
158
159 fprintf('\n=== STEP 9: Generating plots ===\n');
160
161 figure('Color','w','Position',[80 80 1100 850]);
162
163 % -- Plot 1: Irradiance -- full 8 days with forecast overlay
164     ----
165
166 subplot(3,1,1);
167 plot(trainData.Timestamp, trainData.Irradiance, ...
168     'Color', [0.55 0.55 0.55], 'LineWidth', 1.2); hold on;
169 plot(testData.Timestamp, actualIrr, ...
170     'Color', [0.10 0.45 0.85], 'LineWidth', 2);
171 plot(tForecast, irrF, ...
172     'Color', [0.85 0.20 0.10], 'LineWidth', 2, 'LineStyle', '
173     --');
174
175 fill([tForecast; flipud(tForecast)], ...
176     [irrCI_upper; flipud(irrCI_lower)], ...
177     [0.85 0.20 0.10], 'FaceAlpha', 0.15, 'EdgeColor','none');
178
179 xline(trainData.Timestamp(end), '--k', 'Forecast Start', '
180     LabelHorizontalAlignment','right','FontSize',8);
181
182 title('Irradiance -- Training (Days 1-7) + Day 8 Forecast vs
183     Actual','FontSize',11);

```

```

174 ylabel('GHI (W/m^2)'); grid on;
175 legend('Training (Days 1-7)', 'Actual Day 8', 'Forecasted Day 8',
        '95% CI', ...
176         'Location', 'northwest', 'FontSize', 8);
177
178 % -- Plot 2: Temperature
        -----
179 subplot(3,1,2);
180 plot(trainData.Timestamp, trainData.Temperature, ...
181       'Color', [0.55 0.55 0.55], 'LineWidth', 1.2); hold on;
182 plot(testData.Timestamp, actualTemp, ...
183       'Color', [0.10 0.45 0.85], 'LineWidth', 2);
184 plot(tForecast, tempF, ...
185       'Color', [0.85 0.20 0.10], 'LineWidth', 2, 'LineStyle', '
        --');
186 xline(trainData.Timestamp(end), '--k', 'Forecast Start', '
        LabelHorizontalAlignment', 'right', 'FontSize', 8);
187 title('Temperature -- Training (Days 1-7) + Day 8 Forecast vs
        Actual', 'FontSize', 11);
188 ylabel('degC'); grid on;
189 legend('Training (Days 1-7)', 'Actual Day 8', 'Forecasted Day 8',
        'Location', 'northwest', 'FontSize', 8);
190
191 % -- Plot 3: Day 8 zoom -- direct forecast vs actual comparison
        -
192 subplot(3,1,3);
193 yyaxis left;
194 plot(testData.Timestamp, actualIrr, '-o', 'Color', [0.10 0.45
        0.85], ...
195       'LineWidth', 2, 'MarkerSize', 4); hold on;
196 plot(tForecast, irrF, '--s', 'Color', [0.85 0.20 0.10], ...
197       'LineWidth', 2, 'MarkerSize', 4);
198 ylabel('Irradiance (W/m^2)');
199
200 yyaxis right;
201 plot(testData.Timestamp, actualTemp, '-^', 'Color', [0.10 0.65

```

```

0.30], ...
202     'LineWidth',1.2,'MarkerSize',4);
203 plot(tForecast, tempF, '--v','Color',[0.85 0.55 0.10], ...
204     'LineWidth',1.2,'MarkerSize',4);
205 ylabel('Temperature (degC)');
206
207 title(sprintf('Day 8 Zoom -- Actual vs Forecast | Irr R^2 = %.3
    f, Temp R^2 = %.3f', R2_i, R2_t), ...
208     'FontSize',11);
209 xlabel('Hour of Day'); grid on;
210 legend('Actual Irr','Forecast Irr','Actual Temp','Forecast Temp
    ', ...
211     'Location','northwest','FontSize',8);
212
213 sgtitle('PV Digital Twin -- ARIMA Forecast Validation (Hourly,
    Kathmandu)', ...
214     'FontSize', 13, 'FontWeight', 'bold');
215
216 saveas(gcf, 'forecast_validation.png');
217 fprintf('Plot saved: forecast_validation.png\n');
218
219 %% -- DONE
    -----
220 fprintf('\n=====');
221 fprintf(' ALL DONE. Files generated:\n');
222 fprintf(' forecast_vs_actual_day8.csv\n');
223 fprintf(' forecast_metrics.csv\n');
224 fprintf(' forecast_validation.png\n');
225 fprintf('=====');
226 fprintf('\nNext step: Feed forecast_vs_actual_day8.csv into
    your\n');
227 fprintf('Simulink model to get the final POWER forecast for Day
    8.\n');

```

Listing 7.2 SARIMA forecast and validation script.

Appendix D: Arduino Code for LDR Sensor

The Arduino sketch below reads the LDR voltage from analog pin A0 and the DHT11 temperature sensor from digital pin 2. The LDR voltage is converted to irradiance using the calibration coefficients from Section 4.11. Both values are sent over the serial port as a comma-separated pair, which is then read by MATLAB during real-time operation.

```
1  #include <DHT.h>
2
3  // Pins
4  const int    LDR_PIN    = A0;
5  const int    DHT_PIN    = 2;
6  #define      DHT_TYPE    DHT11
7
8  // Constants
9  const float  V_REF      = 5.0;
10 const float  ADC_MAX    = 1023.0;
11
12 // LDR calibration coefficients:  $G = a * V^b$ 
13 const float  a_cal      = 77.23;
14 const float  b_cal      = 3.6631;
15
16 DHT dht(DHT_PIN, DHT_TYPE);
17
18 unsigned long last_temp_read_ms = 0;
19 const unsigned long TEMP_PERIOD_MS = 2000;
20 float last_temp_c = 25.0;
21
22 void setup() {
23     Serial.begin(9600);
24     dht.begin();
25     delay(2000);    // DHT11 stabilisation
26 }
27
28 void loop() {
```

```

29  // LDR -> irradiance
30  int    raw_ldr    = analogRead(LDR_PIN);
31  float  v_ldr      = (raw_ldr / ADC_MAX) * V_REF;
32  float  irradiance = a_cal * pow(v_ldr, b_cal);
33  if (irradiance < 0) irradiance = 0;
34
35  // DHT11 -> temperature (refresh every 2 s)
36  if (millis() - last_temp_read_ms >= TEMP_PERIOD_MS) {
37      float t = dht.readTemperature();
38      if (!isnan(t)) last_temp_c = t;
39      last_temp_read_ms = millis();
40  }
41
42  // Send "irradiance,temperature\n"
43  Serial.print(irradiance, 2);
44  Serial.print(",");
45  Serial.println(last_temp_c, 2);
46
47  delay(1000);
48  }

```

Listing 7.3 Arduino code for LDR + DHT11 acquisition.

D.1 MATLAB Real-Time Interface

The MATLAB-side script below opens the serial port, parses each incoming line, updates the Simulink model's irradiance and temperature inputs, runs the simulation for one sample horizon, and logs the result.

```
1  %% Real-Time PV Digital Twin
2  clc; clear; close all;
3
4  SERIAL_PORT = 'COM6';
5  BAUD_RATE   = 9600;
6  MODEL_NAME  = 'Grid_connected_single_phase_solar';
7  SIM_TIME    = '0.15';
8  DURATION_SEC = 300;
9  LOG_FILE     = 'realtime_log.csv';
10
11  IRR_BLOCK    = [MODEL_NAME '/Constant'];
12  TEMP_BLOCK   = [MODEL_NAME '/Constant1'];
13
14  % Open serial port
15  arduino = serialport(SERIAL_PORT, BAUD_RATE);
16  configureTerminator(arduino, "LF");
17  flush(arduino); pause(2);
18
19  % Open model and prepare logging
20  load_system(MODEL_NAME);
21  fid = fopen(LOG_FILE, 'w');
22  fprintf(fid, 'Timestamp,Irradiance_Wm2,Temperature_C,Power_W\n'
23           ');
24
25  t_start = tic;
26  while toc(t_start) < DURATION_SEC
27      line = readline(arduino);
28      values = str2double(split(line, ','));
29      if numel(values) < 2 || any(isnan(values)), continue; end
30
31      irr = values(1);
```

```

31     temp = values(2);
32
33     set_param(IRR_BLOCK, 'Value', num2str(irr));
34     set_param(TEMP_BLOCK, 'Value', num2str(temp));
35
36     simOut = sim(MODEL_NAME, 'StopTime', SIM_TIME);
37     P       = simOut.power_out;
38     if isobject(P), P = P.Data; end
39     n_steady = max(1, round(length(P) * 0.9));
40     P_now    = max(0, mean(P(n_steady:end)));
41
42     t_now = toc(t_start);
43     fprintf(fid, '%.1f,%.2f,%.2f,%.2f\n', t_now, irr, temp,
44           P_now);
45
46     fclose(fid);
47     clear arduino;

```

Listing 7.4 MATLAB real-time digital twin interface.

Appendix E: PV Module Datasheet (RenewSys DESERV 3M6H-335)

Parameter	Value
Manufacturer	RenewSys India Pvt. Ltd.
Model	DESERV 3M6H-335
Cell technology	Multi-crystalline silicon
Number of cells	72 (6×12)
<i>Electrical parameters at STC ($G = 1000 \text{ W/m}^2$, $T = 25^\circ \text{C}$, $AM 1.5$)</i>	
Rated maximum power, $P_{\max}$	335.0 W
Open-circuit voltage, V_{oc}	46.27 V
Short-circuit current, I_{sc}	9.41 A
Voltage at MPP, V_{mp}	37.9 V
Current at MPP, I_{mp}	8.85 A
Power tolerance	0 to +4.99 W _p
<i>Thermal coefficients</i>	
Temperature coefficient of V_{oc} , β	-0.36099 %/°C
Temperature coefficient of I_{sc} , α	+0.102 %/°C
NOCT (assumed in model)	45 °C
<i>Mechanical and system data</i>	
Module weight	21.2 kg
Maximum system voltage	1500 V (DC)
Output tolerance	0 to +3 %
<i>Certification</i>	
Standards	IEC 61215, IEC 61730, IEC 61701, IEC 62716

Table 7.4 RenewSys DESERV 3M6H-335 module specifications.

The manufacturer's datasheet, including I–V and P–V characteristic curves at multiple irradiance and temperature levels, is available from the product portal.

Appendix F: MATLAB P&O MPPT Controller Code

```
1 function D = Pand0(Param, Enabled, V, I)
2 % MPPT controller based on the Perturb & Observe algorithm.
3 % D output = Reference for DC link voltage (Vdc_ref)
4 % Enabled input = 1 to enable the MPPT controller
5 % V input = PV array terminal voltage (V)
6 % I input = PV array current (A)
7 % Param input:
8 Dinit = Param(1); % Initial value for Vdc_ref
9 Dmax = Param(2); % Maximum value for Vdc_ref
10 Dmin = Param(3); % Minimum value for Vdc_ref
11 deltaD = Param(4); % Increment value used to increase/decrease
    Vdc_ref
12
13 persistent Vold Pold Dold;
14
15 dataType = 'double';
16
17 if isempty(Vold)
18     Vold = 0;
19     Pold = 0;
20     Dold = Dinit;
21 end
22
23 P = V * I;
24 dV = V - Vold;
25 dP = P - Pold;
26
27 if dP ~= 0 && Enabled ~= 0
28
29     if dP < 0
30
31         if dV < 0
32             D = Dold + deltaD;
33         else
34             D = Dold - deltaD;
```

```

35         end
36
37     else
38
39         if dV < 0
40             D = Dold - deltaD;
41         else
42             D = Dold + deltaD;
43         end
44
45     end
46
47 else
48     D = Dold;
49 end
50
51 if D >= Dmax || D <= Dmin
52     D = Dold;
53 end
54
55 Dold = D;
56 Vold = V;
57 Pold = P;
58
59 end

```

Listing 7.5 MATLAB implementation of the Perturb and Observe (P&O) MPPT algorithm.

The implemented P&O controller continuously perturbs the operating voltage and observes the resulting power variation to track the maximum power point (MPP) of the photovoltaic array. The algorithm was integrated into the MATLAB/Simulink PV digital twin model for real-time MPPT operation under varying irradiance and temperature conditions.